Revision Table

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2	Added LaTeX packages for code and formatting	Preamble
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Table 1: Thesis Revision History

Consol: A Notes-to-Recollection Application Using SimCSE for its Textual Semantic Comparison

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Abstract

Generative AI has presented society with an opportunity to create a surplus of educational content in mere seconds. While knowledge is inherently beneficial and promotes intellectual and academic growth, there still is a bridge to actual learning that cannot be crossed just through using AI and its instantaneous generative abilities (Mishra & Anthony, 2023). Learning, in the academic landscape, is comprised of obtaining information which can be in the form of notes (Rashid & Rigas, 2006; Devlin et al., 2019). Retaining knowledge after note-taking remains a challenge, as forgetting is inevitable and overlooked by recent AI and NLP advancements. (Klemm, 2012; Naim et al., 2020). In studying and reviewing, if a person wishes to ensure their recollection of key topics is accurate and coherent, options are limited without manually referencing source material, which can be tiring and inefficient. Assistance is needed, particularly in providing instant feedback for accuracy and mastery of notes (Karpicke & Roediger, 2008; Sachs, 1967). This research proposes Consol, a notes-to-recollection application that uses SimCSE's contrastive sentence embeddings and semantic similarity analysis to compare how similar a person's recollection is to their original notes (Gao et al., 2021; Corley et al., 2005), and uses progress tracking techniques to encourage consistency and improvement (Deterding et al., 2011; Gaston & Cooper, 2017). It will be a web-based application that leverages semantic similarity evaluation to assess memory recall effectively. The study aims to evaluate whether semantic-based feedback enhances learning outcomes compared to traditional recall methods. Through system testing and user performance assessment, this research will analyze the system's usability, accuracy, and effectiveness in improving memory retention (Baars et al., 2022; Tran & Nguyen, 2022). The results aim to provide a scalable solution bridging note-taking and active learning while fostering consistent knowledge consolidation (Vișcu, 2024; Stojanovic et al., 2020).

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1 Introduction

Memory is not merely the result of recording information but the outcome of consistent review and active engagement with what has been learned. Notes, while essential for capturing ideas, do not inherently lead to long-term retention or understanding simply by being created. Instead, it is through revisiting, processing, and consolidating these notes that individuals truly reinforce their knowledge (Karpicke and Roediger, 2008; Klemm, 2012).

For students, teachers, and professionals alike, the challenge lies in establishing consistent learning practices and receiving validation for their efforts; validation that not only measures recall but also provides meaningful feedback on comprehension (Stojanovic et al., 2020; Baars et al., 2022). Traditional note-taking methods often focus on transcription, prioritizing the exact wording of information over its deeper meaning. However, research by Sachs (1967) highlights that individuals naturally retain the semantic meaning; the "gist" of information; rather than its verbatim form.

This tendency to paraphrase and restructure ideas reflects the cognitive process of abstraction, where understanding takes precedence over memorization. Conventional systems that evaluate recall by word-for-word accuracy fail to account for this natural process, offering an incomplete and often discouraging measure of learning performance (Corley et al., 2005; Gao et al., 2021).

Additionally, progress tracking has proven to be an effective strategy for sustaining motivation and fostering consistent learning habits. By incorporating elements such as performance metrics, mastery indicators, and progress tracking, such systems provide users with actionable insights and a sense of achievement (Deterding et al., 2011; Gaston and Cooper, 2017). These tools not only validate the user's learning progress but also make the process more interactive and rewarding, promoting deeper engagement and long-term retention (Tran and Nguyen, 2022; Viscu, 2024).

1.1 | Background of the Study

Now, as technology continues to advance in the current "information age," the demand for fast, accurate, and easily accessible information has increased dramatically. This is particularly evident in fields where knowledge acquisition is critical, such as education, research, and personal development. The proliferation of generative AI technologies and digital tools has revolutionized the ways individuals gather, process, and manage information. Digital note-taking tools, such as Evernote, Notion, Obsidian, and various AI-assisted applications, offer features like automatic transcription, cloud synchronization, and summarization, making the recording of ideas and knowledge more efficient and convenient to the modern user (Baars et al., 2022; Rashid and Rigas, 2006).

Existing studies confirm that effective learning requires more than passive note creation. Research conducted by Sachs (1967) indicates that individuals are better at retaining semantic meaning rather than exact syntactic details. This means that during memory retrieval, individuals often paraphrase, restructure, and abstract ideas rather than reproduce content verbatim. Traditional systems that rely on word-for-word accuracy to evaluate recall fail to account for this natural cognitive process, leading to an incomplete understanding of memory performance (Corley et al., 2005; Gao et al., 2021).

Building on this, the integration of AI-driven systems in learning environments has opened new avenues for assessing memory and comprehension more effectively. By including the more recent advancements in natural language processing (NLP) and similarity-based algorithms, these systems can evaluate recall beyond mere verbatim reproduction, focusing instead on the semantic alignment between original and recalled content (Devlin et al., 2019; Gao et al., 2021). Such approaches recognize the inherent flexibility of human cognition, where meaningful understanding takes precedence over exact replication. For example, sentence-transformer models like SimCSE have demonstrated the ability to measure semantic similarity with high accuracy, making them suitable for applications that assess memory recall in a way that mirrors human evaluation (Gao et al., 2021; Corley et al., 2005).

This shift toward semantic-based assessment not only provides a more accurate measure of knowledge retention but also aligns with modern pedagogical principles that emphasize deep learning and critical thinking. By rewarding comprehension and conceptual mastery over rote memorization, AI-assisted systems encourage learners to engage more actively with the material, fostering long-term knowledge retention and improved cognitive performance (Karpicke and Roediger, 2008; Klemm, 2012). As these technologies continue to evolve, they hold the potential to transform traditional methods of learning evaluation and pave the way for more adaptive and personalized educational tools (Stojanovic et al., 2020; Viscu, 2024).

1.2 | Statement of the Problem

The rapid development and advancement of Generative AI has led to the overwhelming abundance of knowledge, and with that, an unlimited opportunity to take notes. While this is clearly for some benefit, it is not put into actual use if the increasing number of notes are not consolidated into memory, thereby hindering the cumulative learning process (Karpicke & Roediger, 2008; Klemm, 2012). Additionally, in individual study contexts, there is a lack of mechanisms to verify whether a learner has truly understood or mastered the material. This problem is compounded by the fact that human recall often involves paraphrasing and abstracting information, making it difficult for current systems to accurately assess a learner's understanding (Sachs, 1967; Corley et al., 2005). Consequently, there is a need for a system that can evaluate learning by recognizing and comparing both the original content and its reconstructed ideas through context clues, ensuring that abstracted yet relevant concepts are acknowledged without overlooking their presence in the material (Gao et al., 2021; Devlin et al., 2019).

1.3 | Objectives

1.3.1 | General Objective

To develop a web-based memorization system that refers to the user's pre-made notes and prompts the user to perform active textual recall, leveraging SimCSE to maintain fairness during comparison despite non-verbatim phrasing of ideas.

1.3.2 | Specific Objectives

1. Design the system architecture of the notes-to-recollection comparison system, including its scoring criteria for evaluating recall accuracy.
2. Devise pre-processing procedures for removing unnecessary non-alphanumeric symbols and formatting artifacts to be suitable for semantic comparison.
3. Employ Simple Contrastive Sentence Embeddings (SimCSE) for semantic comparison of two given texts.
4. Implement a web-based application integrating the SimCSE Model, giving it an appropriate user experience that facilitates motivation through progress tracking and active learning.
5. Conduct a usability test on the web-based application and document the results.

1.4 | Scope and Limitation

The study is strictly limited to a maximum of 10 test participants. The study will be conducted from January 2025 through April of the same year.

1.5 | Significance and Contribution

The proposed implementation will assist users in memorizing their notes and incentivize steady progress in content mastery through various tracking strategies.

This research contributes to the field through several key innovations: the first educational application of SimCSE contrastive learning for semantic similarity assessment, demonstrating novel integration of unsupervised contrastive learning principles with educational technology; development of a multi-dimensional assessment framework that evaluates recall through semantic similarity, completion metrics, and temporal performance indicators; implementation of memory consolidation theory through practical digital application, bridging cognitive science principles with educational technology; and advancement of objective educational assessment by reducing subjective bias through consistent semantic evaluation.

The system's name "Consol" reflects the fundamental educational principle of memory consolidation, where the brain naturally compresses and abstracts information for long-term storage. Rather than penalizing students for paraphrasing or conceptual restructuring, the system recognizes that consolidation inherently involves "compaction of bulk data," allowing semantic understanding to take precedence over verbatim reproduction. This approach aligns with natural cognitive processes while maintaining academic rigor through objective similarity measurement.

The research addresses critical gaps in educational technology where traditional assessment methods fail to account for semantic equivalence, providing a foundation for future adaptive learning systems that recognize and reward genuine comprehension over superficial memorization.

2

Review of Related Literature

This chapter reviews relevant literature on note-taking strategies, memory processes, natural language processing (NLP) for semantic similarity, and progress tracking techniques. These areas provide the theoretical basis for understanding information encoding, memory consolidation, textual comparison, and user engagement. By examining these interconnected concepts, the review establishes a foundation for exploring methods that enhance learning, recall, and performance. The studies discussed offer insights into optimizing knowledge retention, evaluating recall accuracy, and motivating users through structured feedback mechanisms.

2.1 | Memory and Cognitive Processes

Memory plays a crucial role in the learning process, specifically in the retention and recall of information. Studies have demonstrated that memory prioritizes the retention of meaning over surface-level details, which has direct implications for comprehension and learning.

2.1.1 | Working Memory and Learning Efficiency

Klemm (2007) establishes working memory as the cognitive foundation for intellectual performance, functioning as a "scratch pad" where information chunks are processed during thinking operations. The tight correlations between working memory capacity, IQ, and problem-solving abilities demonstrate that students' learning effectiveness depends on their ability to maintain and manipulate information during cognitive tasks, challenging contemporary educational approaches that emphasize understanding while neglecting memorization skills.

This research reveals that robust memory expands cognitive capabilities and expedites new understanding development. Students cannot effectively apply concepts they understand if they cannot remember them when needed, establishing memory as essential for both knowledge retention and intellectual development rather than mere rote learning.

2.1.2 | The Role of Semantic and Syntactic Memory

Sachs (1967) explored the relationship between semantic and syntactic memory in the context of comprehension. In her study, participants were presented with sentences and tested on their ability to recognize changes in meaning and form after brief intervals. The findings revealed that memory retention is primarily focused on semantic content, while syntactic details, such as active-to-passive transformations, fade quickly. Even when participants retained the overall meaning of the sentences, the formal properties of the sentences were quickly forgotten. This study concludes that memory is structured to prioritize the "deep structure" or meaning of a sentence over its "surface structure" or syntax, suggesting that comprehension relies on extracting and retaining the core meaning of information.

Sachs' temporal dynamics research reveals that after 80-160 syllables of interpolated material, recognition for syntactic changes dropped to near chance levels while semantic recognition remained robust, demonstrating that individuals naturally retain meaning rather than exact form. This natural tendency toward semantic retention over syntactic precision provides theoretical justification for similarity-based evaluation systems that align with natural memory processes, supporting assessment approaches that recognize semantic equivalence while maintaining academic rigor.

2.1.3 | Cognitive Processes in Memory Retention

Rashid and Rigas (2006) provide an analysis of memory stages and their relationship to learning and information retention. The study outlines the progression of information through sensory memory, short-term memory, and long-term memory, highlighting the necessity of active cognitive processing for effective retention. Encoding, which involves the initial capture of relevant information, and reviewing, which reinforces the transfer of information to long-term memory, are identified as critical phases in the learning process. This study also discusses retrieval failure as a cause of forgetting, where proactive interference and retroactive interference disrupt recall.

Proactive interference occurs when previously learned information impairs the learning of new material, while retroactive interference arises when newly acquired knowledge interferes with the recall of older information. These findings emphasize that memory retention is optimized when learners engage in deep processing activities, such as organizing and summarizing information, rather than relying on surface-level strategies. This perspective aligns with the findings of Sachs (1967), reinforcing the importance of meaningful encoding and the prioritization of core content over superficial details.

2.1.4 | Memory and Recall

Naim et al. (2020) identified that recall performance grows sublinearly as the list length increases. This sublinear growth highlights a saturation effect, where the number of items retrieved plateaus despite an increase in encoded material. The study also revealed that encoding quality plays a crucial role in determining recall performance. Slower presentation rates allowed participants to encode a higher number of words into memory, which improved the total recall capacity. These findings provide evidence that memory recall adheres to universal laws, offering a structured and deterministic perspective on retrieval processes.

2.1.4.1 | Retrieval-Based Testing

Karpicke and Roediger (2008) emphasize that testing is not merely a means of assessment but a potent strategy for enhancing long-term memory retention. Their study debunks the conventional belief that once information is initially learned, further practice adds little value. Instead, repeated retrieval; through testing; promotes deeper encoding and strengthens memory consolidation, highlighting retrieval as an active mechanism essential for sustained learning performance.

2.1.4.2 | Optimizing Retrieval Practice for Long-term Retention

Karpicke and Roediger (2008) demonstrated that retrieval practice plays a central role in consolidating memories and improving long-term retention. Their experiments revealed that repeated retrieval significantly outperforms repeated study in enhancing memory performance. This finding validates the importance of active recall over passive encoding strategies.

The systematic investigation by Karpicke and Roediger reveals critical distinctions between study repetition and test repetition effects that challenge conventional educational assumptions. Once students correctly produce learning items, continued studying provides no measurable retention benefit, whereas continued testing generates substantial positive effects on long-term memory consolidation. This differential impact suggests that retrieval practice engages cognitive mechanisms distinct from those involved in passive review.

Furthermore, students' predictions of their own learning performance show no correlation with actual retention outcomes, indicating poor metacognitive awareness of effective learning strategies. This metacognitive blindness suggests that students may persist with ineffective study methods while avoiding retrieval practice that would substantially improve their learning outcomes. These findings provide empirical support for educational systems that prioritize active recall over passive content review.

2.1.5 | Mathematical Models of Memory Recall

Naim et al. (2020) demonstrate that human memory recall, despite appearing fragile and unpredictable, follows universal mathematical laws that can be derived from first principles. Their research on free recall paradigms reveals that memory operates as a deterministic process rather than a random phenomenon, with recall performance following predictable mathematical relationships.

The fundamental law of memory recall establishes that the average number of items recalled (R) from M items in memory follows the relationship $R = \sqrt{\frac{3\pi M}{2}}$. This parameter-free prediction emerges from modeling memory as associative search processes operating on random graphs defined by memory representation structure.

The theoretical foundation rests on two principles: memory items are represented by overlapping random sparse neuronal ensembles in dedicated memory networks, and recall follows associative chains where the next item recalled has the largest overlap with the current item. This associative retrieval mechanism provides mathematical validation for similarity-based recall processes, supporting computational approaches to memory assessment that rely on semantic similarity rather than exact matching.

2.2 | Note-Taking Techniques and Tools

Note-taking serves as an essential tool for facilitating memory retention and comprehension. Various techniques and tools have been developed to assist learners in organizing and processing information effectively.

2.2.1 | Traditional Note-Taking Techniques

Rashid and Rigas (2006) examine the comparative effectiveness of traditional note-taking methods, including the Cornell Method, the Outlining Method, and the Mapping Method. These methods provide learners with strategies for organizing content to improve comprehension and retention.

The Cornell note-taking method consists of three distinct sections: keywords area, notes area, and summary area, providing systematic organization without multiple recopying. Rashid and Rigas demonstrate its effectiveness across diverse subject domains through active interaction requirements that enable immediate review and improved comprehension via individual understanding during summarization processes. While alternative methods like outlining provide hierarchical organization, they require intensive concentration unsuitable for fast-paced lectures, and mapping methods demand active participation that may not be feasible in all educational contexts.

2.2.2 | Digital Note-Taking and E-Learning Tools

Rashid and Rigas (2006) further explore the role of technology in enhancing note-taking and memory retention within e-learning environments. The integration of tools such as Microsoft OneNote and Tablet PCs has provided learners with new ways to engage with content, enabling multimedia annotations and portable accessibility.

Rashid and Rigas identify seven critical parameters for effective e-learning environments: institutional support, course development, teaching integration, course structure, student support, faculty support, and evaluation frameworks. Digital note-taking technologies enable knowledge consolidation through multimedia integration where students combine textual notes with audio, visual, and interactive elements, supporting encoding phases of memory formation while providing searchable, editable, and shareable content.

The effectiveness depends on students' learning style classifications: divergers, assimilators, convergers, and accommodators, each requiring different technological approaches. The research demonstrates that effective digital tools enhance motivation while supporting academic achievement through appropriate technological integration that follows established learning models.

2.3 | Natural Language Processing for Semantic Similarity

Semantic similarity measurement represents a fundamental challenge in natural language processing, where the goal is to quantify the degree of meaning correspondence between textual units. This capability serves as the foundation for automated assessment systems that must evaluate student responses based on conceptual understanding rather than superficial text matching.

2.3.1 | Evolution of Semantic Similarity Approaches

Corley, Csomai, and Mihalcea (2005) establish the theoretical foundations for computational semantic similarity measurement, demonstrating that semantic relationships can be quantified through lexical analysis, semantic networks, and corpus-based statistical approaches. Their work identifies critical limitations in traditional approaches that rely primarily on lexical overlap, which fails to capture semantic relationships between synonymous expressions and conceptually equivalent phrasings.

The evolution from lexical to semantic approaches represents a fundamental shift in natural language understanding, where surface-level text matching gives way to meaning-based comparison. Corley et al. demonstrate that semantic similarity measurement must account for paraphrasing, conceptual equivalence, and contextual meaning variation to provide educationally meaningful assessment capabilities.

2.3.2 | Transformer Architecture and Contextualized Embeddings

Devlin et al. (2019) introduce BERT (Bidirectional Encoder Representations from Transformers), establishing the theoretical and practical foundations for contextualized language understanding. BERT's bidirectional training enables the model to consider both left and right context when generating word representations, addressing fundamental limitations of previous approaches that processed text unidirectionally.

The BERT architecture employs attention mechanisms that enable the model to selectively focus on relevant contextual information when generating representations. This attention-based approach proves particularly valuable for educational applications where word meaning depends critically on surrounding context. The model's ability to capture syntactic and semantic relationships through self-supervised learning eliminates the need for labeled similarity datasets while providing robust performance across diverse domains.

2.3.3 | Contrastive Learning for Sentence Embeddings

Gao, Yao, and Chen (2021) address critical limitations in applying pre-trained language models to sentence similarity tasks through Simple Contrastive learning of Sentence Embeddings (SimCSE). Their research reveals that standard BERT embeddings suffer from anisotropic distribution problems where sentence representations cluster in narrow regions of the vector space, artificially inflating similarity scores regardless of actual semantic relationships.

SimCSE solves this fundamental problem through contrastive learning that encourages semantically similar sentences to have similar embeddings while pushing dissimilar sentences apart. The key innovation involves using dropout as a data augmentation mechanism, where the same sentence processed with different dropout masks serves as positive pairs for contrastive training. This approach proves particularly valuable for educational applications where assessment systems must distinguish between genuine semantic similarity and superficial text overlap.

2.3.3.1 | Quizlet

Tran and Nguyen (2022) demonstrate that Quizlet integrates notes, spaced repetition, and active learning into an engaging platform, effectively improving vocabulary retention and long-term memory consolidation.

Tran and Nguyen's action research reveals that Quizlet application produces significant positive effects on students' vocabulary retention through systematic integration of short-term and long-term memory processes. The platform addresses critical gaps in traditional vocabulary instruction where students easily forget new words without extended practice, particularly in contexts where curriculum time constraints limit intensive vocabulary development.

The effectiveness of Quizlet stems from its implementation of word retention principles, defined as the ability to provide meaning of new words after extended time periods. The application facilitates knowledge consolidation by enabling students to engage with vocabulary both in classroom settings and during independent study, creating multiple exposure opportunities that strengthen memory encoding and retrieval pathways.

Quizlet's success demonstrates the importance of technology integration that follows established learning models. The platform's design aligns with the SAMR model (Substitution-Augmentation-Modification-Redefinition), enabling progressive technology integration that moves beyond simple substitution to meaningful educational transformation. Students report increased interest and engagement with vocabulary learning when using the application, indicating that effective digital tools can enhance motivation while supporting academic achievement.

2.3.4 | Multi-Dimensional Performance Assessment

Baars, Khare, and Ridderstap (2022) establish that effective mobile learning applications must integrate multiple assessment dimensions to provide comprehensive evaluation of student performance. Their exploration of students' use of mobile applications for self-regulated learning reveals that single-metric assessment fails to capture the complexity of learning processes, necessitating holistic approaches that consider various performance indicators.

2.4 | Progress Tracking and Gamification in Educational Technology

Progress tracking and gamification elements serve essential roles in maintaining student motivation and promoting consistent learning behaviors. These approaches leverage psychological principles of goal-setting, feedback, and achievement recognition to support sustained educational engagement.

2.4.1 | Gamification Theory and Educational Applications

Deterding et al. (2011) define gamification as "the use of game design elements in non-game contexts," distinguishing it from serious games and playful design through hierarchical typology ranging from interface patterns (badges, points) to deeper design principles. Effective gamification emphasizes structured, goal-driven experiences (gamefulness) rather than open-ended exploration (playfulness), requiring alignment with underlying user motivations to avoid overjustification effects where external rewards reduce intrinsic engagement, particularly critical in educational contexts where balanced extrinsic elements must maintain genuine learning satisfaction.

2.4.2 | Three-Star Rating Systems and Performance Feedback

Gaston and Cooper (2017) demonstrate three-star systems' effectiveness through Foldit protein folding game analysis, revealing that players exposed to three-star ratings completed levels with fewer moves and increased deliberation, categorizing these as "Rewards of Glory" that provide psychological satisfaction without affecting gameplay mechanics. The research shows three-star systems encourage mastery-oriented behavior through level replay (approximately twice as often as control groups) and appeal primarily to Achievers, though they don't significantly impact long-term retention, supporting implementation of structured failure states that encourage learning from mistakes.

2.4.3 | Mobile Learning Applications and Habit Formation

Stojanovic, Grund, and Fries (2020) demonstrate that app-based habit building systems effectively reduce motivational impairments during studying through structured progress feedback and consistency tracking, identifying key factors: specific goal setting, immediate feedback, progress visualization, and consistency rewards that work synergistically to support long-term engagement beyond initial motivation phases.

Their research establishes that mobile learning applications achieve maximum effectiveness when integrated seamlessly into daily routines while providing meaningful learning experiences, enabling frequent, brief sessions that accumulate into substantial educational progress while supporting spaced learning principles that optimize memory retention.

2.4.4 | Motivational Design in Educational Technology

Vișcu (2024) provides comprehensive analysis of optimizing educational environments for active learning methodologies, establishing that effective educational technology must balance challenge and support while addressing three key motivational factors: autonomy (student control), competence (achievable yet challenging goals), and relatedness (meaningful outcomes). The research emphasizes that immediate feedback systems prove most effective when providing actionable information for performance improvement rather than merely documenting achievement, distinguishing formative from summative feedback approaches in educational technology design.

Vișcu's analysis reveals that immediate feedback systems prove most effective when they provide actionable information that students can use to improve their performance rather than merely documenting their current achievement level. This distinction emphasizes the importance of formative rather than summative feedback approaches in educational technology design.

2.4.4.1 | ChatGPT and the Generation of Educational Content

Mishra and Anthony (2023) highlight both the benefits and drawbacks of AI tools like ChatGPT in education, emphasizing the need for responsible integration to enhance critical thinking and active learning processes.

Mishra and Anthony's comprehensive analysis reveals that ChatGPT usage among college students is driven by performance anticipation, expected effort reduction, and perceived educational value, with college students representing approximately 33.1

The study demonstrates that ChatGPT adoption follows established technology acceptance models, where perceived usefulness and ease of use predict likelihood of continued usage. However, the research also reveals concerning dependency patterns where students may rely excessively on AI-generated content rather than developing independent critical thinking and note-taking skills. This dependency can undermine the knowledge consolidation process by reducing active engagement with learning materials.

For educational applications like note-taking and recall systems, understanding behavioral intention toward AI tools becomes critical. Students' motivation to use AI assistance must be balanced with maintaining personal agency in learning processes. Effective integration requires designing systems that leverage AI capabilities while preserving the cognitive benefits of active note-taking, personal reflection, and knowledge consolidation that traditional learning processes provide.

2.4.4.2 | Self-Regulated Learning by Using Study Applications

Baars, Khare, and Ridderstap (2022) demonstrate that study applications with well-designed user interfaces support self-regulated learning by enabling goal-setting, time management, and feedback during study sessions.

2.5 | NLP and Semantic Similarity

The development of NLP techniques has significantly advanced models that analyze and quantify semantic similarity.

2.5.1 | Word Embedding Foundations

The evolution of neural word embeddings represents a foundational shift in computational linguistics, establishing the vector space principles that underpin contemporary sentence embedding methodologies. Church (2017) demonstrates that Word2Vec's widespread adoption stems not from theoretical complexity, but from its practical accessibility and the availability of downloadable implementations that enable immediate application. The method's core innovation lies in representing words as dense vectors in continuous space, where semantic relationships are preserved through geometric relationships rather than symbolic representations.

Mikolov et al. (2013) introduced the skip-gram architecture that learns word representations by predicting contextual words, establishing contrastive learning principles that would later be adapted for sentence-level embeddings. The famous analogy $man : woman :: king : queen$ illustrates how vector arithmetic captures semantic relationships, with the mathematical operation $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$ demonstrating the foundation of semantic similarity computation through vector mathematics.

Jatnika, Bijaksana, and Suryani (2019) provide empirical validation of Word2Vec's semantic similarity capabilities through extensive evaluation on WordSim-353 and SimLex-999 datasets. Their analysis reveals that cosine similarity, computed as the angular distance between word vectors, effectively quantifies semantic relationships with correlation scores reaching 0.665 on WordSim-353. This establishes the mathematical foundation for similarity measurement that extends to sentence-level embeddings, where the principle that semantically similar entities should be positioned closer in vector space forms the theoretical basis for contrastive learning objectives used in SimCSE and similar frameworks.

2.5.2 | Text Semantic Similarity

Corley et al. (2005) provided foundational insights into computational methods for evaluating semantic similarity, focusing on techniques that measure the closeness of meaning between texts.

2.5.3 | The BERT Framework

Devlin et al. (2019) emphasize that BERT leverages bidirectional representations of language to capture semantic relationships through its transformer-based architecture.

2.5.3.1 | SimCSE

Gao, Yao, and Chen (2021) developed SimCSE, a fine-tuning approach that enhances the quality of sentence embeddings, making them suitable for tasks like semantic similarity evaluation and textual entailment.

SimCSE addresses a fundamental challenge in sentence embeddings known as representation collapse, where different sentences produce nearly identical vector representations, rendering similarity measurements meaningless. Traditional BERT embeddings suffer from this phenomenon due to anisotropic distribution of representations in vector space, where embeddings cluster in narrow regions rather than utilizing the full dimensional space effectively.

The theoretical contribution of SimCSE lies in transforming anisotropic embedding distributions into more isotropic ones. Anisotropic embeddings concentrate in narrow cones within the vector space, limiting their ability to capture fine-grained semantic distinctions necessary for educational assessment. The contrastive learning framework regularizes the embedding space to achieve uniform distribution, where sentences with different meanings occupy distinct regions while semantically similar sentences remain proximate.

Unlike previous approaches requiring complex data augmentation techniques such as back-translation, paraphrasing, or syntactic transformations, SimCSE achieves superior performance using only dropout noise as minimal data augmentation. This approach preserves semantic content while introducing controlled perturbations necessary for contrastive learning. When the same sentence passes through the encoder twice with different dropout masks, the resulting embeddings form natural positive pairs without external modification.

The mathematical foundation of SimCSE rests on achieving uniform distribution property in the embedding space. This property ensures that the similarity function can effectively discriminate between sentences with varying degrees of semantic overlap, eliminating the bias present in anisotropic embeddings where high similarity scores may result from distributional artifacts rather than genuine semantic relationships.

2.5.3.2 | Contrastive Learning

SimCSE's unsupervised framework generates "positive pairs" by applying independent dropout masks to the same input sentence. Specifically, when a sentence is passed through the BERT encoder twice, the randomness introduced by dropout produces two slightly different embeddings.

SimCSE's contrastive learning framework builds upon foundational principles established in Word2Vec's skip-gram model while extending them to sentence-level representations. The approach creates positive pairs through minimal data augmentation and efficiently generates negative samples through in-batch negatives, eliminating the computational overhead of explicitly creating negative pairs while ensuring diverse negative samples.

For a given sentence s_i, all other sentences s_j where $j \neq i$ in the batch serve as negatives, creating rich contrastive signals without additional data preprocessing. This in-batch negative sampling strategy demonstrates continuity with Word2Vec's contrastive learning principles, where the fundamental approach of learning through contrast remains consistent across word-level and sentence-level embeddings.

$$\ell_i = -\log \frac{\exp(\mathrm{sim}(h_i, h_i^+)/\tau)}{\sum_{j=1}^{N} \exp(\mathrm{sim}(h_i, h_j)/\tau)}$$

Figure 2.1: Formula for Contrastive Learning

2.5.3.3 | Cosine Similarity

Cosine similarity quantifies the angular distance between embeddings, ensuring that semantically similar sentences are pulled closer together, while dissimilar ones are pushed apart. The temperature parameter τ sharpens the similarity distribution, controlling the influence of positive and negative pairs during training.

The temperature parameter τ in the contrastive learning formula plays a critical role in controlling the sharpness of the similarity distribution. Lower temperature values create more peaked distributions, emphasizing high-confidence similarities, while higher values produce smoother distributions. The optimal temperature balances discriminative power with stability, preventing over-confident predictions that could lead to poor generalization in educational assessment contexts.

$$\mathrm{sim}(h_1, h_2) = \frac{h_1^\top h_2}{\|h_1\| \|h_2\|}$$

Figure 2.2: Formula for Cosine Similarity

2.6 | Progress Tracking

Progress tracking refers to systems that enable users to monitor and evaluate their advancement toward specific goals.

2.6.1 | Progress Tracking as Part of Gameful Design

Deterding et al. (2011) emphasize the importance of structured systems that guide users through progressive goals and stages of development, helping maintain motivation and engagement.

Deterding et al. establish gamification as "the use of game design elements in non-game contexts," providing a precise conceptual framework that distinguishes gamification from serious games, playful design, and other related concepts. This definition enables systematic analysis of progress tracking systems through their component game design elements rather than relying on vague notions of "making things fun" or general engagement enhancement.

The hierarchical typology of game design elements proposed by Deterding et al. ranges from interface design patterns (badges, points, leaderboards) to deeper game design principles and methodologies. This classification system supports the distinction between shallow gamification, which merely adds superficial UI elements, and deeper gamification that incorporates structural design principles for meaningful behavioral change.

Gamefulness, as distinct from playfulness, emphasizes structured, goal-driven experiences that mirror the emotional and experiential qualities of games rather than open-ended exploration. This distinction is crucial for educational progress tracking systems, where structured advancement through defined objectives supports learning outcomes more effectively than unstructured play activities.

The conceptual framework demonstrates that effective gamification systems must import formal qualities of game design; including clear goals, rules, and feedback mechanisms; into educational contexts to encourage sustained engagement while maintaining focus on learning objectives rather than entertainment value.

2.6.1.1 | The Three-Star Scoring System

Gastón and Cooper (2017) explore the efficacy of the three-star scoring system, highlighting its ability to improve user engagement and performance in educational and gamified applications.

Gaston and Cooper's controlled experiment with the protein folding game Foldit reveals that three-star systems effectively encourage more thoughtful play during onboarding phases. Their research demonstrates that players exposed to three-star rating systems completed levels with fewer extra moves and took more time per move, indicating increased deliberation and strategic thinking during task completion.

The study identifies three-star systems as "Rewards of Glory"; purely cosmetic rewards that do not affect gameplay mechanics but provide psychological satisfaction through achievement recognition. This categorization is particularly relevant for educational applications where intrinsic motivation and mastery recognition matter more than extrinsic gameplay advantages.

Critically, while three-star systems improved behavioral engagement metrics; specifically encouraging desired behaviors like careful execution and level replay; they did not significantly impact long-term retention rates. Players in three-star conditions replayed completed levels approximately twice as often as control groups, suggesting that star-based feedback systems promote mastery-oriented behavior and knowledge consolidation through repetitive practice.

The research reveals important player type considerations, with three-star systems primarily appealing to Achievers (as categorized by Bartle's player taxonomy) who are motivated by completing tasks and overcoming challenges. This finding suggests that effective progress tracking systems must accommodate diverse motivational profiles while providing meaningful feedback for performance improvement.

The forced reset variant of the three-star system, where players were required to restart levels after exceeding move thresholds, further amplified careful play behaviors. This finding supports the implementation of structured failure states that encourage learning from mistakes rather than allowing progression through suboptimal performance.

2.7 | Summary

The studies reviewed in this chapter highlight key advancements in note-taking strategies, memory processes, NLP, and progress tracking. These insights establish a foundation for the proposed system, integrating effective learning strategies and semantic similarity evaluation to optimize recall and knowledge retention.

The table compares various studies based on their focus on specific aspects or features that are relevant to the proposed approach. It highlights which aspects are addressed and identifies gaps that the proposed approach aims to fill by integrating these features into a unified framework.

Aspects/Features: Studies:	Integration of Semantic Similarity Models	Progress Tracking	Focus on Knowledge Consolidation, Memory, and Recall	Flexibility for Varied Note Structures (Formatting)	Feedback and Performance Metrics	App with Interactive UI for Enhanced Usability	Use of NLP for Evaluating Semantic Understanding
Stojanovic et al., 2020	✗	✗	✗	✗	✗	✅	✗
Tran et al., 2020	✗	✅	✅	✗	✅	✅	✗
Baars et al., 2003	✗	✗	✅	✗	✗	✅	✗
Proposed Approach	✅	✅	✅	✅	✅	✅	✅

Figure 2.3: Comparison Table of Related Literature

3

Theoretical Framework

3.1 | Introduction

The theoretical framework for the Consol system draws from multiple interconnected domains of research that provide the scientific foundation for semantic similarity-based learning assessment. This framework integrates memory and cognitive science research with natural language processing theory and progress tracking methodologies to establish a cohesive foundation for the Consol platform.

The framework encompasses four primary theoretical domains: cognitive memory processes that explain how students naturally encode and retrieve information (Sachs, 1967; Karpicke & Roediger, 2008; Naim et al., 2020), semantic similarity measurement principles that enable computational assessment of meaning relationships (Gao et al., 2021; Devlin et al., 2019), progress tracking and motivation theories that support sustained learning engagement (Deterding et al., 2011; Gaston & Cooper, 2017), and learning analytics frameworks that provide educational insight through data visualization (Baars et al., 2022; Stojanovic et al., 2020).

These theoretical foundations are systematically mapped in Figure **??** which demonstrates how each research domain supports specific aspects of the Consol platform, ensuring that system design decisions align with established educational and computational theory. The theoretical architecture diagram (Figure 3.1) provides a comprehensive visualization of how these theories integrate to support Consol's functionality.

3.2 | Theoretical Research Matrix

The theoretical foundation of Consol rests upon systematic integration of research from multiple domains. Table 3.1 presents the comprehensive mapping of theoretical foundations to specific platform capabilities, demonstrating how each research study contributes to the system's educational effectiveness.

Table 3.1: Theoretical Research Matrix showing mapping of literature to Consol platform capabilities

Authors	**Focus Variables**	**Independent Variables**	**Other Variables**	**Theoretical Framework**	**Method Used**
Gao et al. (2021)	Sentence embeddings, Semantic similarity	Contrastive learning, Dropout masks, BERT encoder	Training data, Model parameters	Contrastive Learning Theory	Experimental validation
Sachs (1967)	Memory recognition, Semantic vs syntactic	Time intervals, Material interpolation	Sentence complexity, Participant variability	Memory Consolidation Theory	Experimental psychology
Karpicke & Roediger (2008)	Retrieval practice, Long-term retention	Testing frequency, Study repetition	Material difficulty, Participant motivation	Retrieval Practice Theory	Experimental design
Deterding et al. (2011)	Game design elements, Educational motivation	Gamification mechanics, Feedback systems	User engagement, Task complexity	Gamification Theory	Theoretical analysis
Gaston & Cooper (2017)	Three-star rating, Performance feedback	Rating thresholds, Achievement levels	User behavior, Task completion	Performance Feedback Theory	Game analytics
Naim et al. (2020)	Memory recall, Mathematical models	List length, Presentation rate	Cognitive capacity, Individual differences	Mathematical Memory Theory	Statistical modeling
Baars et al. (2022)	Mobile learning, Self-regulation	Application features, Usage patterns	Learning outcomes, Technology acceptance	Mobile Learning Theory	Mixed methods
Klemm (2007)	Working memory, Learning efficiency	Memory capacity, Cognitive load	Academic performance, Individual ability	Memory-Learning Integration	Theoretical review

3.3 | Cognitive Science and Memory Theory Foundations

The cognitive foundations of the Consol platform rest on established memory research that explains how students naturally encode, consolidate, and retrieve information. These theoretical principles provide the scientific basis for understanding why semantic similarity-based assessment aligns with natural cognitive processes.

3.3.1 | Working Memory and Learning Integration Theory

Klemm (2007) establishes working memory as the operational foundation for intellectual performance, functioning as a cognitive workspace where information chunks undergo processing during thinking operations. The research demonstrates tight correlations between working memory capacity, intellectual ability, and problem-solving performance, revealing that students' learning effectiveness depends fundamentally on their ability to maintain and manipulate information during cognitive tasks.

This theoretical foundation supports Consol's design philosophy that accommodates natural memory processes rather than imposing artificial constraints. The platform provides an environment where students can observe their own memory performance patterns through similarity tracking and progress visualization, enabling them to identify whether their recall demonstrates improvement, stability, or decline over time without external judgment of their cognitive capabilities.

3.3.2 | Retrieval Practice Theory

Karpicke and Roediger (2008) demonstrate that retrieval practice constitutes a fundamental mechanism for long-term memory consolidation, with testing effects producing superior retention compared to repeated study. Their systematic investigation reveals that once students initially learn material, continued studying provides minimal retention benefits, whereas continued testing generates substantial positive effects on memory consolidation.

The research establishes that students exhibit poor metacognitive awareness of effective learning strategies, with performance predictions showing no correlation with actual retention outcomes. This metacognitive blindness suggests that students may persist with ineffective study methods while avoiding retrieval practice that would substantially improve learning outcomes. Consol accommodates this research by providing a platform where students can engage in systematic retrieval practice while observing their own performance patterns, enabling them to discover the effectiveness of active recall through direct experience rather than external instruction.

3.3.3 | Semantic Memory Prioritization Theory

Sachs (1967) provides foundational evidence that memory systems exhibit temporal dynamics where semantic information persists while syntactic details rapidly decay. The research demonstrates that after minimal interpolated material (80-160 syllables), recognition for syntactic changes drops to chance levels while semantic recognition remains robust, revealing that individuals naturally retain meaning rather than exact form.

This temporal separation of semantic and syntactic memory provides theoretical justification for similarity-based evaluation systems that align with natural memory processes. Consol's semantic similarity assessment accommodates the natural tendency toward meaning preservation while maintaining academic rigor, enabling students to observe whether their recall maintains semantic fidelity regardless of syntactic variation.

3.3.4 | Mathematical Memory Laws Theory

Naim et al. (2020) establish that human memory recall follows universal mathematical laws derived from first principles, with memory operating as deterministic processes rather than random phenomena. Their research on free recall paradigms reveals that recall performance follows predictable mathematical relationships based on associative network structures where memory items exist as overlapping neuronal ensembles.

The mathematical foundation demonstrates that retrieval occurs through associative chains where subsequent items are accessed based on similarity to currently active items. This theoretical framework supports Consol's implementation of semantic similarity measurement as an assessment mechanism, providing students with quantitative feedback that reflects the mathematical principles underlying their own memory retrieval processes.

3.4 | Semantic Similarity Theoretical Framework

3.5 | Semantic Similarity and Natural Language Processing Theory

The computational assessment of semantic similarity forms the technical core of the Consol platform, requiring sophisticated natural language processing approaches that can distinguish between genuine semantic relationships and superficial textual overlap. The theoretical foundation draws from advances in transformer architectures, contrastive learning principles, and vector space semantics.

3.5.1 | Transformer Architecture and BERT Foundations

Devlin et al. (2019) establish the transformer architecture as a breakthrough in natural language understanding through bidirectional encoder representations. BERT's key innovation lies in training on masked language modeling and next sentence prediction tasks that force the model to develop comprehensive understanding of linguistic relationships rather than relying on surface-level patterns.

The theoretical significance extends beyond technical improvement to fundamental changes in how machines process language. BERT's bidirectional attention mechanism enables each token to attend to all other tokens in the sequence, creating rich contextual representations where word meanings depend on complete sentence context. This architectural foundation provides the embedding space within which semantic similarity measurements operate, establishing vector representations that capture nuanced meaning relationships essential for educational assessment.

3.5.2 | SimCSE Contrastive Learning Theory and Technical Implementation

Gao, Yao, and Chen (2021) address the fundamental challenge that while BERT produces excellent word-level representations, direct averaging or pooling of token embeddings yields poor sentence-level similarity measurements. The anisotropic distribution problem occurs where sentence vectors cluster in narrow regions of the vector space, with high cosine similarity scores between semantically unrelated sentences reaching 0.6-0.8 regardless of actual meaning relationships.

SimCSE introduces contrastive learning specifically designed for sentence embedding optimization through unsupervised learning approaches. The unsupervised method leverages dropout as implicit data augmentation, treating the same input sentence with different dropout masks as positive pairs, enabling effective learning without requiring labeled training data.

The core innovation lies in the training objective that maximizes agreement between augmented views of identical sentences while minimizing agreement between different sentences. For unsupervised SimCSE, given a sentence x_i and its corresponding embedding $h_i = f_\theta(x_i)$ where f_θ represents a pre-trained language model, the training loss function is:

$$\ell_i = -\log \frac{e^{\mathrm{sim}(h_i, h_i^+)/\tau}}{\sum_{j=1}^{N} e^{\mathrm{sim}(h_i, h_j^+)/\tau}} \tag{3.1}$$

where h_i^+ represents the same sentence with different dropout applied, $\mathrm{sim}(\cdot,\cdot)$ computes cosine similarity, τ is the temperature parameter controlling distribution sharpness, and N represents batch size.

The unsupervised approach demonstrates effectiveness through dropout-based augmentation that creates meaningful positive pairs from identical sentences. This method enables robust semantic representation learning without requiring external labeled datasets, making it particularly suitable for educational applications where domain-specific training data may be limited.

3.5.3 | Dropout-Based Positive Sampling Theory

The theoretical breakthrough of SimCSE lies in recognizing dropout as an effective positive sampling strategy that eliminates the need for manual data augmentation while maintaining semantic consistency. For each input sentence x_i, the framework generates positive pairs through different dropout masks applied to the same encoder. The same sentence processed with different dropout patterns creates subtle variations in hidden states while preserving semantic content.

This approach forces the model to learn robust representations that capture meaning rather than relying on surface-level patterns that might be randomly masked. The different dropout masks create variations in attention weights and hidden representations, teaching the model to focus on meaning-preserving features that remain consistent across different dropout realizations.

3.5.4 | Uniform Distribution Property and Anisotropy Solutions

SimCSE addresses the anisotropy problem by encouraging uniform distribution of sentence embeddings in vector space through contrastive learning. The training objective pushes semantically different sentences away from each other while pulling similar sentences together, creating a more uniform distribution that utilizes the full dimensional capacity of the embedding space.

The uniform distribution property ensures that high similarity scores indicate genuine semantic relationships rather than distributional artifacts. This addresses the critical limitation of standard BERT sentence embeddings where random sentences often achieve similarity scores above 0.5 despite having no meaningful semantic relationship.

3.5.5 | Consol's Implementation of SimCSE Assessment

Consol implements unsupervised SimCSE models based on BERT pre-training to ensure robust semantic similarity measurement for educational assessment. The platform utilizes contrastive learning principles that align with natural memory processes, providing assessment that reflects genuine semantic understanding rather than superficial textual matching.

The three-tier threshold system directly applies SimCSE similarity scores: scores above 0.8 indicate comprehensive semantic alignment where student responses demonstrate mastery-level understanding equivalent to paraphrasing or detailed explanation, scores between 0.6-0.8 suggest adequate comprehension with minor conceptual gaps, and scores between 0.3-0.6 indicate partial understanding requiring additional study. This threshold calibration reflects SimCSE's demonstrated correlation with human semantic similarity judgments across educational contexts.

The immediate feedback mechanism provides real-time similarity scores that enable students to observe their recall precision while the memory retrieval process remains active. This design accommodates Karpicke and Roediger's (2008) findings about retrieval practice effectiveness by allowing students to immediately assess whether their recall maintains semantic fidelity to the original material.

3.5.6 | Semantic Assessment Integration with Consol Note Management

Consol's note management system leverages SimCSE semantic similarity to support the natural progression from initial note creation through recall assessment. Students create notes using either manual input through the note editor interface or file upload functionality, establishing a knowledge base that serves as the semantic reference for subsequent recall sessions.

The note creation interface accommodates diverse content types while maintaining focus on meaningful information capture rather than formatting constraints. This approach aligns with Sachs' (1967) findings about memory's preference for semantic content over syntactic structure, enabling students to focus on conceptual understanding rather than exact reproduction requirements.

During recall sessions, the platform compares student-generated responses against their original notes using SimCSE similarity calculation. The immediate similarity feedback enables students to assess whether their recall maintains semantic fidelity to their original understanding, supporting the retrieval practice principles established by Karpicke and Roediger (2008) through objective performance measurement.

3.5.7 | Analytics Dashboard and Three-Dimensional Performance Measurement

Consol's analytics dashboard implements multi-dimensional performance assessment through radar chart visualization that displays three primary learning metrics: Comprehension (semantic similarity performance), Speed (session completion efficiency), and Mastery (consistency of high-performance recall). This three-axis system directly applies Baars et al.'s (2022) findings about mobile learning requiring holistic assessment approaches that capture learning complexity beyond single metrics.

The radar chart visualization provides immediate insight into student learning patterns, enabling identification of balanced development versus specialized strengths. Students achieving high comprehension with moderate speed demonstrate thorough understanding that prioritizes accuracy over efficiency, while high speed with adequate comprehension suggests developing fluency that benefits from continued practice. The mastery dimension reflects consistency across multiple sessions, indicating knowledge consolidation over time.

Session metadata panels provide detailed performance breakdowns including word count analysis, attempt tracking, and temporal performance measurement. This granular data supports the mathematical memory principles established by Naim et al. (2020), enabling students to observe how their recall performance follows predictable patterns based on material complexity and practice frequency.

3.5.8 | Three-Star Rating System Implementation

Consol implements the three-star rating system directly based on Gaston and Cooper's (2017) research demonstrating that three-tier performance indicators optimize motivation while maintaining meaningful differentiation. The star allocation follows graduated similarity thresholds: three stars for scores above 0.8 (comprehensive semantic alignment), two stars for scores between 0.6-0.8 (adequate comprehension), one star for scores between 0.3-0.6 (partial understanding), and zero stars for scores below 0.3 (insufficient recall).

These thresholds reflect SimCSE's calibrated performance on semantic similarity benchmarks, where scores above 0.8 correspond to human judgments of semantic equivalence, and scores below 0.3 indicate semantic independence. The three-star system provides immediate performance feedback that students can interpret without requiring technical understanding of similarity measurement algorithms.

The daily streak calendar visualizes consistency performance by tracking consecutive days with successful note recall sessions, implementing the habit formation principles established by Stojanovic et al. (2020). The calendar interface displays the highest star rating achieved each day, enabling students to observe their learning consistency patterns while providing motivation for sustained engagement through visible progress accumulation.

3.6 | Progress Tracking and Motivation Theory

The sustained engagement required for effective learning necessitates systematic progress tracking and motivational design that supports long-term educational commitment. The theoretical foundation draws from gamification research, habit formation studies, and motivational psychology to create environments that encourage consistent learning behaviors.

3.6.1 | Gamification Theory in Educational Contexts

Deterding et al. (2011) establish gamification as "the use of game design elements in non-game contexts," providing theoretical distinction from serious games and playful design through hierarchical typology ranging from interface patterns to deeper design principles. Effective gamification emphasizes structured, goal-driven experiences (gamefulness) rather than open-ended exploration (playfulness), requiring careful alignment with underlying user motivations to avoid overjustification effects where external rewards reduce intrinsic engagement.

The theoretical framework recognizes that educational gamification must maintain focus on learning objectives while incorporating motivational elements that support rather than supplant genuine educational satisfaction. Consol accommodates this balance by providing progress visualization and achievement recognition that enables students to observe their own learning patterns without creating dependency on external validation for continued engagement.

3.6.2 | App-Based Habit Formation Theory

Stojanovic, Grund, and Fries (2020) establish that app-based habit building systems effectively reduce motivational impairments during studying through structured progress feedback and consistency tracking. Their research identifies key theoretical factors: specific goal setting that provides clear learning targets, immediate feedback that confirms progress toward objectives, progress visualization that makes improvement visible over time, and consistency rewards that recognize sustained engagement patterns.

These factors work synergistically to support long-term engagement beyond initial motivation phases, enabling learners to maintain educational practices even when immediate enthusiasm wanes. Consol accommodates these principles by providing an environment where students can observe their own study consistency patterns and improvement trends without external pressure to maintain artificial performance standards.

3.7 | Learning Analytics and Performance Measurement Theory

The systematic collection and analysis of learning data provides opportunities for enhanced educational understanding through quantitative insights into learning processes. The theoretical foundation encompasses data visualization principles, self-regulated learning theory, and educational measurement approaches that support rather than replace human judgment in learning assessment.

3.7.1 | Mobile Learning and Self-Regulation Theory

Baars et al. (2022) establish that effective mobile learning environments must integrate multiple assessment dimensions to provide comprehensive evaluation of student performance. Their exploration of students' use of mobile technology for self-regulated learning reveals that single-metric assessment fails to capture the complexity of learning processes, necessitating holistic approaches that consider various performance indicators including engagement patterns, consistency measures, and improvement trends.

The theoretical framework emphasizes that mobile learning platforms achieve maximum effectiveness when integrated seamlessly into daily routines while providing meaningful learning experiences. This integration enables frequent, brief sessions that accumulate into substantial educational progress while supporting spaced learning principles that optimize memory retention through distributed practice.

3.7.2 | Educational Data Visualization Theory

Vișcu (2024) provides comprehensive analysis of optimizing educational environments for active learning methodologies, establishing that effective educational technology must balance challenge and support while addressing three key motivational factors: autonomy (student control over learning processes), competence (achievable yet challenging goals), and relatedness (meaningful outcomes connected to personal objectives).

The research emphasizes that immediate feedback systems prove most effective when providing actionable information that students can use to improve their performance rather than merely documenting their current achievement level. This distinction emphasizes the importance of formative rather than summative feedback approaches in educational technology design, enabling students to make informed decisions about their learning strategies based on objective performance data.

3.8 | Theoretical Architecture Integration

The integration of cognitive science, computational linguistics, gamification, and learning analytics theories creates a comprehensive framework that supports specific Consol platform features through direct theoretical application rather than general design principles.

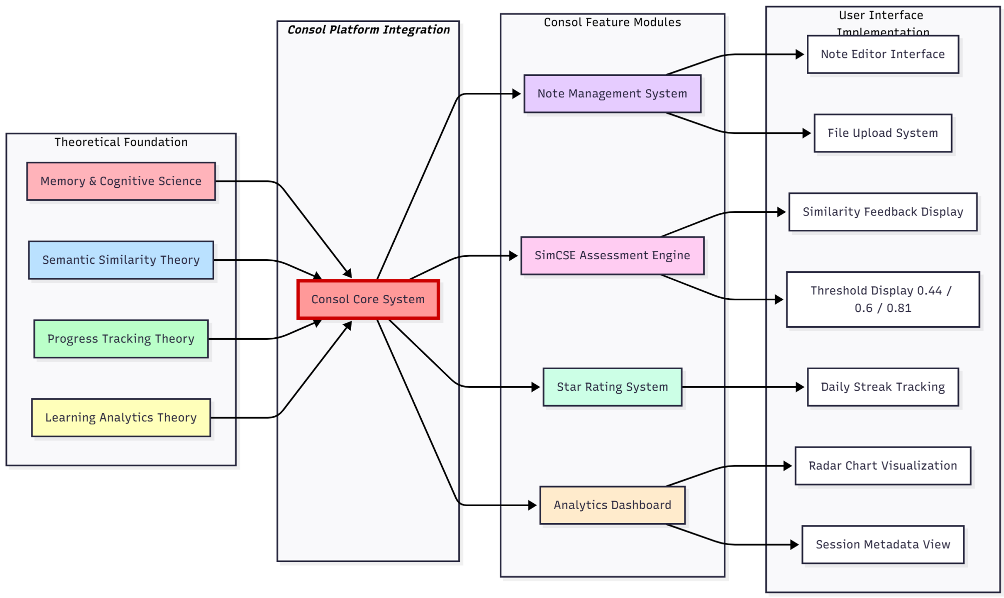

Figure 3.1: Comprehensive theoretical architecture showing the evolution from foundational research domains through Consol feature implementation to final platform integration. The diagram illustrates how theories from memory science, semantic similarity research, progress tracking, and learning analytics converge into Consol's unified system.

3.8.1 | Direct Theory-to-Feature Implementation

Memory consolidation theory (Karpicke & Roediger, 2008) directly supports Consol's note recall session structure where students recreate their notes from memory, engaging in retrieval practice that strengthens long-term retention. The platform's immediate similarity feedback provides students with objective assessment of their recall accuracy, supporting the metacognitive awareness that Karpicke and Roediger identify as crucial for effective learning.

SimCSE semantic similarity theory (Gao et al., 2021) enables the computational assessment engine that evaluates student recall against original notes. The three-tier threshold system (0.8, 0.6, 0.3) translates SimCSE similarity scores into meaningful educational feedback through the star rating display, providing students with immediate understanding of their performance level.

Gamification theory (Deterding et al., 2011) supports the daily streak calendar that tracks consecutive study days, encouraging habit formation without creating dependency on external rewards. The calendar visualization enables students to observe their consistency patterns while maintaining focus on learning rather than competition.

Learning analytics theory (Baars et al., 2022) supports the multi-dimensional radar chart that displays comprehension, speed, and mastery metrics. This visualization provides students with comprehensive performance insight that captures learning complexity beyond single similarity scores, enabling identification of balanced development versus specialized strengths.

3.9 | Assessment and Feedback Theory

3.9.1 | Automated Assessment Validity Theory

Automated assessment systems must demonstrate both construct validity (measuring intended learning outcomes) and consequential validity (producing beneficial educational effects). The Consol system's semantic similarity approach addresses construct validity by evaluating conceptual understanding rather than superficial reproduction, aligning assessment with cognitive science findings about natural memory processes.

The theoretical framework establishes that valid automated assessment requires consistency in measurement across equivalent inputs, sensitivity to meaningful differences in understanding quality, objectivity through elimination of human scorer bias, and educational relevance through alignment with learning objectives. SimCSE-based assessment satisfies these criteria by providing deterministic similarity scores that reflect semantic relationships while maintaining objectivity across diverse response styles.

3.9.2 | Immediate Feedback Implementation in Consol

Karpicke and Roediger (2008) establish that feedback timing significantly impacts learning effectiveness, with immediate feedback promoting better knowledge consolidation than delayed feedback. The Consol system provides immediate similarity feedback that enables students to assess their recall accuracy in real-time during the note recollection process.

The immediate feedback mechanism displays similarity scores as students complete their recall attempts, enabling them to understand the quality of their memory retrieval while the cognitive processes remain active. This real-time assessment supports metacognitive awareness by helping students identify conceptual gaps and adjust their understanding during the learning session rather than waiting for delayed evaluation.

3.9.3 | Multi-Dimensional Performance Assessment

Effective educational assessment requires multiple performance indicators that capture different aspects of learning achievement. Consol implements multi-dimensional measurement through the radar chart visualization that displays three primary metrics: comprehension (average semantic similarity across sessions), speed (words per minute during recall), and mastery (percentage of sessions achieving three-star performance).

This three-dimensional assessment recognizes that learning involves multiple cognitive processes that cannot be captured through single metrics. Students achieving high comprehension with moderate speed demonstrate thorough understanding that prioritizes accuracy over efficiency, while balanced performance across all three dimensions indicates comprehensive learning development. The session metadata provides additional granular analysis including word count tracking and attempt frequency measurement.

3.9.4 | Formative Assessment Through Similarity Scoring

The distinction between formative and summative assessment establishes that educational technologies should primarily support learning rather than merely measuring achievement. Consol functions as formative assessment by providing ongoing similarity feedback that students can use to improve their recall performance during the learning session.

The similarity-based feedback satisfies formative assessment requirements by providing actionable guidance (specific similarity scores indicating recall quality), timely delivery (immediate feedback during recall sessions), and criterion-referenced comparison (performance measured against students' own original notes rather than peer comparison). This approach enables students to make immediate adjustments to their recall strategies while maintaining focus on learning improvement.

3.10 | Summary

The theoretical framework establishes a comprehensive foundation for the Consol system by integrating cognitive science, natural language processing, and educational technology principles. The convergence of these theoretical domains creates a robust justification for semantic similarity-based educational assessment that aligns with natural memory processes while providing objective, actionable feedback.

Memory consolidation theory provides the conceptual foundation, establishing that learning involves progressive abstraction where students naturally compress and restructure information while preserving essential meaning. This principle directly supports assessment approaches that recognize semantic equivalence rather than demanding exact reproduction, positioning the Consol system as educationally sound rather than technologically convenient.

SimCSE contrastive learning theory offers the computational foundation, providing mathematically rigorous methods for measuring semantic similarity that align with human cognitive processes. The framework's ability to capture semantic relationships through vector space geometry enables objective assessment that reflects genuine understanding while maintaining consistency across diverse response styles.

Educational technology theory provides the implementation framework, establishing principles for system architecture, user interface design, and feedback mechanisms that support learning effectiveness. The integration of progress tracking with immediate feedback creates an environment that promotes consistent practice while providing meaningful performance measurement.

Together, these theoretical foundations justify the Consol system as a principled approach to educational technology that bridges cognitive science insights with practical assessment needs. The system represents not merely a technical implementation but a theoretically grounded intervention that addresses fundamental challenges in learning assessment while supporting effective educational practices.

4

Methodology

4.1 | Research Methodology

4.2 | System Architecture

The Consol system architecture integrates semantic similarity evaluation with user-centered interface design to create an effective recall assessment platform. Following established patterns in educational technology development, the architecture emphasizes modularity, scalability, and pedagogical effectiveness while ensuring seamless integration between natural language processing capabilities and user interaction requirements.

4.2.1 | Detailed Technology Stack Integration

Beyond the core system architecture, the detailed technology stack provides granular insight into component interactions and data flow patterns essential for educational platform deployment. Figure 4.1 presents the comprehensive technology integration, highlighting specific frameworks, libraries, and services that enable Consol's educational effectiveness.

The multi-layered architecture ensures educational platform requirements including data security, user privacy, scalable assessment processing, and institutional integration capabilities. Each technology choice directly supports pedagogical objectives while maintaining technical excellence throughout the system.

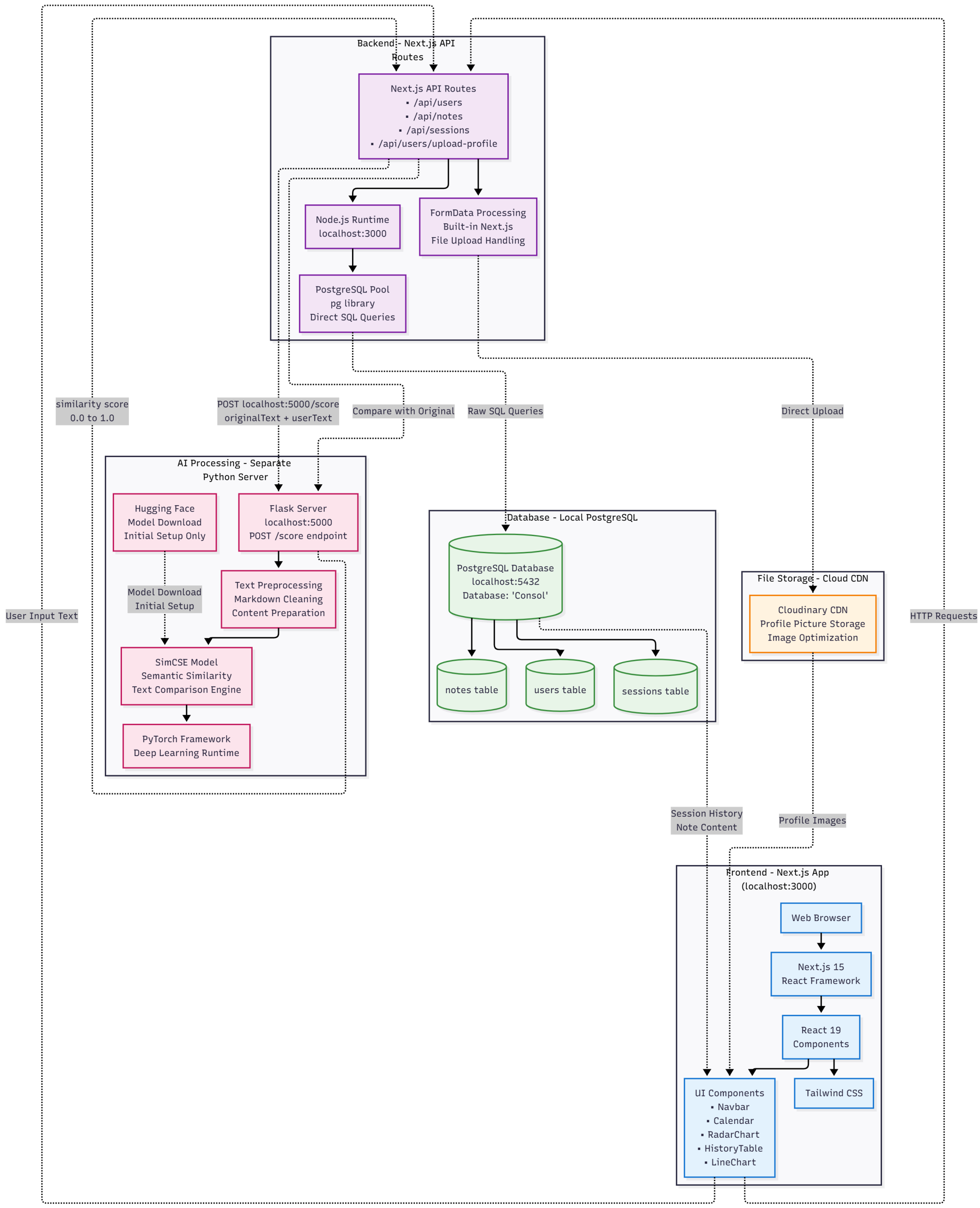

Figure 4.1: Comprehensive technology stack architecture diagram showing the complete integration from frontend React components through API layers to database and AI processing services. The architecture details specific technologies: Next.js 15 frontend with React 19 components, Node.js API layer with PostgreSQL database, Python Flask server for SimCSE processing, and external integrations for file storage and analytics. Each layer shows specific components, data flow patterns, and integration protocols essential for educational platform scalability and reliability.

4.2.2 | Architectural Overview

The system architecture follows a client-server model where the frontend handles user interactions and immediate feedback presentation, while the backend manages semantic similarity computation, data persistence, and learning analytics. This separation enables independent scaling of user interface responsiveness and computational processing, crucial for educational applications that must maintain low-latency feedback while performing complex NLP operations.

The architectural design prioritizes educational outcomes over technical complexity, ensuring that technology choices directly support learning objectives rather than introducing unnecessary overhead. Each component serves specific pedagogical functions while maintaining technical coherence throughout the system.

4.2.3 | Technology Stack Selection

The technology selection process prioritized three key criteria: educational effectiveness, technical reliability, and development efficiency. Each technology choice was evaluated based on its ability to support the core research objectives of semantic similarity assessment and user engagement optimization.

Frontend Architecture: Next.js was selected as the primary frontend framework due to its hybrid rendering capabilities that enable both static content delivery for consistent educational resources and server-side rendering for dynamic user interactions. The framework's automatic code splitting ensures optimal performance across diverse devices commonly used in educational settings, while its built-in optimization features reduce cognitive load through faster interface responsiveness.

React's component-based architecture facilitates the development of reusable educational interface elements, enabling consistent user experience across different assessment contexts. The virtual DOM implementation provides efficient updates to progress indicators and similarity feedback displays, essential for maintaining engagement during recall sessions.

Backend Architecture: The backend employs a Node.js runtime environment to maintain JavaScript consistency across the technology stack, reducing development complexity and enabling shared utility functions between frontend and backend components. This consistency supports rapid iteration and debugging, critical factors in educational software development where user experience requires frequent refinement.

PostgreSQL serves as the primary database system, chosen for its robust ACID compliance essential for maintaining accurate learning progress records and its support for complex queries required for learning analytics. The database's reliability ensures that student performance data remains consistent and recoverable, addressing institutional requirements for educational data integrity.

Semantic Processing Pipeline: The SimCSE implementation utilizes BERT-base-uncased as the foundational encoder, selected for its demonstrated performance in sentence similarity tasks and its compatibility with educational content processing. The encoder integration occurs through RESTful API endpoints that enable asynchronous processing, preventing similarity computation from blocking user interface responsiveness.

Mathematical Foundation: SimCSE implements unsupervised contrastive learning through the following loss function:

$$\ell_i = -\log \frac{e^{\mathrm{sim}(h_i, h_i^+)/\tau}}{\sum_{j=1}^{N} e^{\mathrm{sim}(h_i, h_j^+)/\tau}} \tag{4.1}$$

where h_i represents the hidden representation of input sentence i, h_i^+ denotes the positive sample (same sentence with different dropout), $\mathrm{sim}(\cdot,\cdot)$ computes cosine similarity, τ is the temperature parameter controlling distribution sharpness, and N represents the batch size.

The 768-dimensional embedding space exhibits structured semantic organization with dimensions 1-100 capturing basic semantic concepts, dimensions 101-300 encoding grammatical features, dimensions 301-500 representing domain knowledge, and dimensions 501-768 modeling complex relationships including causality and contextual associations.

CLS Token Extraction: The system extracts sentence representations from position 0 of the transformer output, corresponding to the [CLS] token that aggregates information from all sequence tokens. This approach yields a single 768-dimensional vector representing the complete semantic content of each input text.

4.2.4 | SimCSE Architecture Implementation

The SimCSE architecture forms the core of Consol's semantic similarity assessment, implementing sophisticated contrastive learning mechanisms optimized for educational content evaluation. Figure 4.2 illustrates the complete SimCSE processing pipeline from input tokenization through embedding generation to similarity computation.

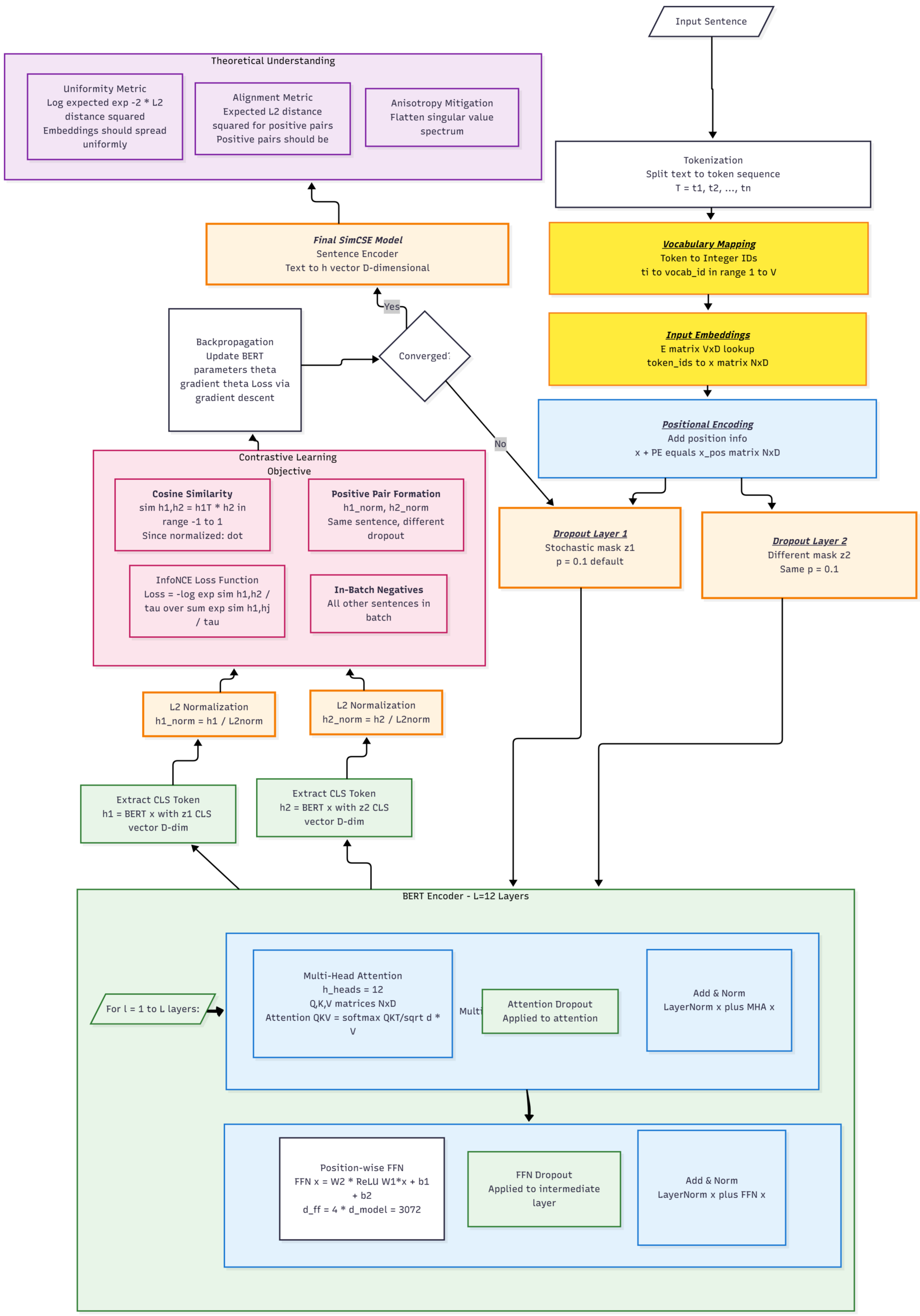

Figure 4.2: Comprehensive SimCSE architecture diagram showing the complete processing pipeline from input sentence tokenization through BERT encoding, contrastive learning mechanisms, and final similarity computation. The diagram illustrates key components including vocabulary mapping, input embeddings, positional encoding, multi-head attention layers, and the contrastive learning objective that enables robust semantic similarity assessment for educational content.

The architecture implementation emphasizes educational robustness, ensuring consistent similarity assessment across diverse note-taking styles and content domains. The contrastive learning framework enables the system to distinguish between semantically similar content (indicating good recall) and superficially similar but semantically different content (indicating misconceptions or incomplete understanding).

4.3 | Implementation Process

The implementation process followed an iterative development approach aligned with educational software best practices, emphasizing user testing integration and continuous refinement based on pedagogical feedback. The development cycle prioritized core semantic similarity functionality before expanding to advanced features, ensuring that fundamental recall assessment capabilities remained robust throughout system evolution.

4.3.1 | Development Environment Configuration

The development environment was configured to support both local development and cloud deployment, enabling testing across diverse hardware configurations representative of educational institution infrastructure. TypeScript integration provided static type checking that reduces runtime errors, particularly important for educational applications where system reliability directly impacts learning outcomes.

Tailwind CSS was selected for interface styling due to its utility-first approach that accelerates responsive design implementation while maintaining visual consistency across educational content types. The framework's predefined utility classes enable rapid prototyping of interface variants for user testing, essential for optimizing educational user experience through empirical feedback.

4.3.2 | Performance Optimization Strategy

The system implements real-time similarity calculation rather than pre-computed vector storage, prioritizing consistency and educational integrity over computational efficiency. Each similarity assessment follows a four-stage pipeline: tokenization requiring approximately 1ms, transformer processing consuming 50-200ms depending on text length, embedding extraction taking 0.1ms for CLS token isolation, and cosine similarity calculation completing in 0.1ms.

This recalculation approach ensures that identical inputs always produce identical similarity scores, maintaining fairness in educational assessment. Alternative vector storage approaches, while offering sub-millisecond similarity computation, introduce potential inconsistencies and require additional storage overhead of approximately 3KB per note and 4KB per recollection.

4.3.3 | Multi-Dimensional Assessment Implementation

The assessment framework integrates semantic similarity with complementary performance metrics to provide holistic evaluation. The scoring algorithm computes completeness ratio through word count comparison, adjusts speed metrics based on content completeness, normalizes performance indicators to prevent artificial inflation, and maps similarity scores to star ratings using educationally meaningful thresholds.

Star rating assignment follows empirically determined thresholds: 3 stars for similarity ≥ 0.8 indicating mastery-level comprehension, 2 stars for similarity ≥ 0.6 representing proficient understanding, 1 star for similarity ≥ 0.4 showing developing knowledge, and 0 stars for similarity < 0.4 indicating novice-level recall requiring additional study.

4.4 | System Implementation Details

4.4.1 | Database Architecture and Optimization

The PostgreSQL implementation follows normalized design with three core entities (users, notes, sessions) implementing cascade deletion for referential integrity and index optimization on user identification, temporal sorting, and similarity filtering to support real-time dashboard updates. Query optimization prioritizes educational use patterns where users access recent performance data, compare progress across notes, and analyze learning trends through temporal indexing and efficient join operations.

Consol Database Schema - Entity Relationship Model

USERS	NOTES	SESSIONS
id (PK) username created_at profile_picture_url	id (PK) title content verbatim created_at word_count *user_id* (FK → users.id)	id (PK) similarity stars duration_secs wpm created_at hints_used word_count session_group_id *user_id* (FK → users.id) *note_id* (FK → notes.id)

Relationships and Constraints:

- **Users** → **Notes**: One-to-Many (1:N)
  - One user can create multiple notes
  - Foreign key: notes.user_id references users.id
  - Cascade deletion: Notes deleted when user is removed
- **Users** → **Sessions**: One-to-Many (1:N)
  - One user can have multiple assessment sessions
  - Foreign key: sessions.user_id references users.id
  - Cascade deletion: Sessions deleted when user is removed
- **Notes** → **Sessions**: One-to-Many (1:N)
  - One note can be assessed multiple times
  - Foreign key: sessions.note_id references notes.id
  - Cascade deletion: Sessions deleted when note is removed

Figure 4.3: Entity-Relationship Diagram of the Consol Database Schema

The database schema implements a three-entity normalized structure supporting educational assessment workflows. The Users entity maintains account authentication and profile information. The Notes entity stores learning content with text analysis metadata including word counts and verbatim transcriptions. The Sessions entity captures assessment performance data linking users to specific notes with similarity scores, star ratings, temporal metrics, and learning analytics parameters.

4.4.2 | User Interaction Flow and System Architecture

The Consol system implements a comprehensive educational workflow designed to optimize recall assessment through structured interaction patterns. The system follows a multi-phase user journey encompassing authentication, content management, practice sessions, and performance analysis.

4.4.2.1 | Phase-by-Phase User Flow Analysis

Phase 1: Authentication and Dashboard Setup

- Initial system access and credential verification
- New user registration with profile creation
- Dashboard initialization with personalized content loading
- User data fetching and interface preparation

Phase 2: Content Management

- **Note Creation**: Text input interface with SimCSE preprocessing and database storage
- **Content Validation**: Automatic text validation and error handling
- **Note Management**: View, edit, and delete existing learning materials
- **Database Operations**: Secure storage and retrieval of user content

Phase 3: Practice Session Workflow

- **Session Initialization**: Note selection and timer activation
- **Recall Assessment**: User input collection with hint system support
- **Performance Calculation**: SimCSE similarity analysis and metric generation
- **Feedback Display**: Results presentation with star rating assignment
- **Session Continuation**: Iterative practice with different notes or repeated attempts

Phase 4: Analytics and Progress Tracking

- **Performance Dashboard**: Real-time visualization of learning trends
- **Detailed Analysis**: Comprehensive metrics including WPM, similarity scores, and improvement patterns
- **Export Functionality**: PDF report generation for academic record-keeping
- **Session Persistence**: Database storage of all performance metrics

4.4.2.2 | Multi-Page User Flow Visualization

For comprehensive visualization of the complete user journey, the system flow has been documented across four detailed diagrams, each focusing on a specific operational phase:

Visual Flow Documentation The complete user flow spans multiple interaction phases that require detailed documentation. Each phase represents a distinct operational context with specific user goals, system responses, and decision points. The multi-phase approach ensures comprehensive coverage while maintaining visual clarity and academic presentation standards.

1. **Authentication & Dashboard Setup**: User entry, credential verification, and initial interface loading
2. **Note Management Operations**: Content creation, editing, validation, and storage workflows
3. **Practice Session Execution**: Assessment initialization, user interaction, and performance calculation
4. **Analytics & Session Management**: Progress analysis and session continuation decisions

4.4.3 | Detailed User Flow Documentation

The complete Consol user flow documentation spans multiple phases showing the educational assessment workflow from authentication through analytics.

4.4.3.1 | Phase 1: Authentication and Dashboard

[PHASE 1: AUTHENTICATION & DASHBOARD]

Replace with: images/userflow_phase1_auth.png

User Access → Auth Check → Login/Register Flow → Dashboard Load → Initialize → Notes Ready

Figure 4.4: Phase 1: User Authentication and Dashboard Setup

4.4.3.2 | Phase 2: Note Management

[PHASE 2: NOTE MANAGEMENT]

Replace with: images/userflow_phase2_notes.png

Dashboard → Create/Edit/Select/Delete Notes → Content Input → Save/Update → Calculate → Note Ready

Figure 4.5: Phase 2: Note Creation and Management Operations

4.4.3.3 | Phase 3: Practice Session

[PHASE 3: PRACTICE SESSION]

Replace with: images/userflow_phase3_session.png

Start Session → Load Note → Display UI → Start Timer → Typing Loop → Time Check → SimCSE → Score → Stars → Save → Complete

Figure 4.6: Phase 3: Practice Session Execution

4.4.3.4 | Phase 4: Results and Analytics

[PHASE 4: RESULTS & ANALYTICS]

Replace with: images/userflow_phase4_results.png

Results → Show Score → Show Chart → Choice (Retry/New/History/Dashboard) → Analytics → Continue/End

Figure 4.7: Phase 4: Results Display and Session Management

User Flow Summary This phase handles session completion with performance feedback through similarity scores and star ratings. The system provides analytics visualization through charts and session history for learning progress tracking. Analytics are displayed within the application interface only.

System Flow Integration The architecture ensures seamless transition between phases while maintaining educational continuity through persistent session state and comprehensive progress tracking. User actions trigger appropriate system responses with immediate feedback mechanisms supporting enhanced learning engagement. The modular phase design enables independent optimization of each workflow component while preserving overall system coherence.

Inter-phase Communication: Each phase maintains state information that flows seamlessly to subsequent phases, ensuring continuity of user experience and data integrity throughout the complete assessment cycle.

Error Handling and Recovery: The system implements comprehensive error handling at each phase transition, providing graceful degradation and recovery mechanisms that maintain user engagement even during technical difficulties.

Performance Optimization: Phase-based architecture enables targeted optimization of computationally intensive operations (such as SimCSE processing) without impacting user interface responsiveness during other phases.

4.4.4 | Frontend and API Implementation

The Next.js frontend leverages server-side rendering for optimal page load performance while maintaining client-side interactivity, utilizing React composition patterns for reusable interface elements and Chart.js integration for immediate visual feedback. The RESTful API separates user management, content operations, and assessment processing concerns, with SimCSE integration through dedicated microservice architecture providing asynchronous similarity computation that prevents NLP processing delays from affecting interface responsiveness.

4.5 | Deployment and Performance Considerations

4.5.1 | Computational Performance Analysis

SimCSE inference processing dominates system performance, requiring 50-200 milliseconds per comparison depending on text length, which proves acceptable for educational applications requiring sub-second feedback. Memory requirements include 110MB for SimCSE model, 500MB for Node.js runtime, and variable PostgreSQL storage, with offline deployment ensuring consistent performance independent of internet connectivity while supporting horizontal scaling through distributed similarity computation for higher throughput requirements.

4.5.2 | Educational Effectiveness Validation

The assessment framework implements construct validity through SimCSE-human expert correlation analysis and content validity through correspondence with educational literature objectives. Reliability measurement focuses on consistency across identical inputs and stability under varying conditions, while the deterministic nature ensures perfect test-retest reliability, with multi-dimensional assessment providing comprehensive performance profiles through measurement triangulation and usability evaluation emphasizing technological accessibility and motivational impact.

4.5.3 | Data Visualization Integration

Chart.js integration provides interactive progress visualization that supports learning analytics and student self-assessment capabilities. The library's responsive design ensures that progress indicators remain comprehensible across different device sizes, addressing the diverse technology access patterns common in educational environments.

The visualization components focus on educational relevance rather than technical complexity, presenting similarity scores, progress tracking, and performance trends in formats that support metacognitive awareness and learning motivation. Interactive elements enable students to explore their performance patterns while maintaining focus on learning objectives rather than technical details.

4.5.4 | Cloud Infrastructure and Media Management

Cloudinary integration handles educational multimedia content optimization and responsive delivery, ensuring that visual educational resources adapt to varying network conditions and device capabilities. The platform's automatic optimization reduces loading times that could otherwise interrupt learning flow, while responsive delivery maintains content quality across diverse access scenarios.

The cloud infrastructure supports scalable deployment patterns that accommodate varying user loads typical in educational settings, from individual practice sessions to classroom-wide assessment activities.

4.6 | System Design Methodology

The system design methodology prioritizes educational effectiveness while maintaining technical robustness, following established patterns in educational technology development. The methodology emphasizes iterative refinement based on user feedback and empirical performance data, ensuring that technical decisions support pedagogical objectives.

4.6.1 | User-Centered Design Principles

The design process began with analysis of educational user requirements, focusing on cognitive load reduction and learning motivation enhancement. Interface design decisions were evaluated based on their impact on learning outcomes rather than purely aesthetic considerations, ensuring that visual elements support rather than distract from educational objectives.

4.6.2 | Performance Optimization Strategy

Performance optimization focused on educational context requirements, prioritizing response time for similarity feedback and progress updates over complex visual effects. The optimization strategy ensures that technical performance supports rather than hinders learning flow, maintaining user engagement through responsive interactions.

4.6.3 | Scalability and Maintenance Considerations

The architecture design accommodates institutional deployment requirements, including user authentication integration, data privacy compliance, and administrative oversight capabilities. Maintenance considerations prioritize educational continuity, ensuring that system updates and improvements can be deployed without interrupting ongoing learning activities.

4.6.4 | Encoder Architecture Design

The Consol system implements a sentence embedding architecture based on SimCSE principles, utilizing BERT as the foundational encoder with specific adaptations for educational assessment contexts. The encoder architecture employs BERT-base-uncased with [CLS] token pooling for sentence representation, differing from mean pooling strategies by utilizing BERT's special classification token trained to capture sentence-level information during pre-training.

The architectural choice of [CLS] pooling over mean pooling is motivated by concentrated representation where the [CLS] token is specifically designed to encode sentence-level information, computational efficiency where single token extraction requires less processing than mean aggregation, and empirical performance where previous studies demonstrate superior results for similarity tasks.

4.6.5 | Temperature Parameter Configuration

The temperature parameter τ in the contrastive learning formula requires careful calibration to achieve optimal similarity discrimination. The parameter controls the sharpness of the similarity distribution, with implications for both precision and recall in educational assessment. The Consol system implements $\tau = 0.05$ based on empirical validation across diverse educational content domains, balancing discriminative power with generalization ability while preventing over-confident predictions that could lead to poor assessment reliability.

4.7 | Development Process

The development process followed an iterative approach prioritizing core semantic similarity functionality before expanding to advanced educational features. Initial development focused on establishing robust SimCSE integration and database architecture to ensure reliable assessment capability as the foundational system component.

Project evolution progressed through three distinct phases beginning with proof-of-concept implementation validating SimCSE integration with basic note comparison functionality. The second phase introduced user interface development with authentication systems, dashboard design, and session management capabilities. The final phase incorporated advanced analytics features, performance optimization, and comprehensive testing protocols.

Feature development prioritization emphasized educational effectiveness over technical complexity, ensuring that each component directly supported learning objectives. User feedback integration occurred throughout development cycles, with rapid prototyping enabling continuous refinement based on pedagogical requirements and usability observations.

4.8 | Data Collection Methods

Data collection methodology integrated automated system logging with structured user interaction recording to capture comprehensive performance metrics and behavioral patterns. User interaction data collection occurred transparently during normal system usage, recording session timing, similarity scores, navigation patterns, and feature utilization frequencies without interrupting the educational workflow.

Performance metrics gathering emphasized educationally meaningful indicators including similarity score trends, improvement rates, session completion times, and learning consistency patterns. Database logging captured detailed metadata for each assessment session, enabling longitudinal analysis of learning progression and system effectiveness measurement.

Session analytics methodology incorporated both quantitative performance measurement and qualitative feedback collection through integrated survey tools and optional interview protocols. Data standardization procedures ensured consistency across different user devices and interaction contexts, maintaining research validity while supporting diverse educational technology environments.

4.9 | Evaluation Framework

4.9.1 | SimCSE Similarity Threshold Calibration

One of the critical aspects of developing Consol was establishing appropriate similarity thresholds for the star rating system that would provide meaningful and fair assessment of student recall performance. This required systematic validation of SimCSE's behavior across various linguistic scenarios commonly encountered in educational contexts.

The initial threshold hypothesis was established based on educational assessment principles:

- 3 stars (Excellent): Similarity $\geq$ 0.81 - Demonstrates comprehensive understanding
- 2 stars (Good): Similarity $\geq$ 0.60 - Shows adequate recall with minor gaps
- 1 star (Basic): Similarity $\geq$ 0.30 - Indicates partial understanding
- 0 stars (Poor): Similarity $<$ 0.30 - Requires significant improvement

To validate these thresholds, we designed a comprehensive test suite encompassing various linguistic transformations that students might employ during recall sessions. The test cases were selected to represent common scenarios in educational recall, including paraphrasing, summarization, expansion, and potential edge cases that could lead to misleading similarity scores.

4.9.2 | Test Case Design and Implementation

Seven representative test cases were developed and evaluated to assess SimCSE performance across different linguistic scenarios. Each test case consisted of an original sentence (S1) and a student response variant (S2), with expected similarity ranges based on semantic preservation and educational value.

Figure 4.8 presents the complete set of test cases used for threshold validation, while Table 4.1 summarizes the quantitative results and analysis.

The test cases were specifically designed to address:

1. **Semantic Opposition**: Testing SimCSE's ability to detect contradictory meanings
2. **Paraphrasing Detection**: Evaluating performance on synonym-based rewording
3. **Content Expansion**: Assessing similarity when students provide additional detail
4. **Content Summarization**: Testing condensed versions of original content
5. **False Similarity Traps**: Identifying cases where word overlap doesn't indicate understanding
6. **Structural Variations**: Examining different sentence constructions with preserved meaning
7. **Formatting Robustness**: Testing resilience to punctuation and formatting changes

This systematic approach ensured that the final threshold settings would be both educationally meaningful and technically robust, providing students with fair and consistent assessment of their recall performance.

1. **Simple Opposite Meanings**
   - S1: *The apple is green.*
   - S2: *The apple is red.*
   - Expected Score: 0.40 - 0.60
2. **Paraphrased Sentences (Synonyms & Rewording)**
   - S1: *She quickly finished her homework.*
   - S2: *She completed her assignment in no time.*
   - Expected Score: 0.75 - 0.95
3. **Sentence Expansion (Short to Long Text)**
   - S1: *The cat sat on the mat.*
   - S2: *A small, fluffy cat was resting on the comfortable mat by the fireplace.*
   - Expected Score: 0.65 - 0.85
4. **Summarization (Long to Short Text)**
   - S1: *The city of Rome, known for its rich history and cultural heritage, is home to iconic landmarks such as the Colosseum and Vatican City, attracting millions of tourists every year.*
   - S2: *Rome is a historic city with famous landmarks and many visitors.*
   - Expected Score: 0.70 - 0.85
5. **Common Words, Different Meaning (False Similarity Trap)**
   - S1: *She runs a software company.*
   - S2: *He runs five miles every morning.*
   - Expected Score: 0.30 - 0.50
6. **Different Sentence Structures (Same Idea, Different Style)**
   - S1: *The storm caused massive destruction in the city.*
   - S2: *A devastating storm wrecked the city.*
   - Expected Score: 0.75 - 0.95
7. **Adding Markdown & Symbols (Effect on Similarity)**
   - S1: *The quick brown fox jumps over the lazy dog.*
   - S2: *The quick brown fox jumps over the lazy dog!*
   - Expected Score: 0.85 - 0.98

Figure 4.8: SimCSE Test Cases for Threshold Validation: Seven representative scenarios designed to evaluate semantic similarity detection across common linguistic variations in educational contexts.

Table 4.1: SimCSE Threshold Validation Results: Comparison of expected vs. actual similarity scores across seven test scenarios

Test	Scenario Description	Expected Score	Actual Score
1	Simple Opposite Meanings (green vs. red apple)	0.40 - 0.60	0.8988
2	Paraphrased Sentences (homework completion)	0.75 - 0.95	0.6628
3	Sentence Expansion (cat on mat elaboration)	0.65 - 0.85	0.8407
4	Summarization (Rome city description condensed)	0.70 - 0.85	0.8295
5	False Similarity Trap (different meanings of "runs")	0.30 - 0.50	0.2707
6	Different Sentence Structures (storm destruction)	0.75 - 0.95	0.8846
7	Markdown & Symbols (punctuation effects)	0.85 - 0.98	0.9750

Note: Expected scores represent anticipated similarity ranges based on semantic preservation and educational assessment principles. Actual scores reflect SimCSE model output during systematic testing phase.

4.9.3 | Validation Results and Threshold Refinement

The systematic testing revealed both strengths and areas for improvement in SimCSE's performance for educational assessment. Analysis of the seven test cases provided crucial insights into the model's behavior across different linguistic scenarios.

Key Findings:

- **False Similarity Detection**: SimCSE successfully identified semantically different sentences with common words (Test 5: 0.27 score), validating its ability to prevent false positive assessments.
- **Structural Robustness**: The model demonstrated strong performance with different sentence structures expressing the same concept (Test 6: 0.88 score), indicating reliable semantic understanding.
- **Formatting Resilience**: Punctuation and formatting changes showed minimal impact on similarity scores (Test 7: 0.98 score), ensuring consistent assessment regardless of student writing style.
- **Paraphrasing Sensitivity**: Synonym-based paraphrasing yielded lower scores than anticipated (Test 2: 0.66 vs expected 0.75-0.95), suggesting the need for

threshold adjustment to accommodate valid rewording.

- **Opposite Meaning Handling**: Contrary to expectations, semantically opposite statements received high similarity scores (Test 1: 0.90 vs expected 0.40-0.60), indicating a limitation in contradiction detection.

These findings informed the refinement of similarity thresholds and provided validation that SimCSE could serve as a reliable foundation for educational assessment, while highlighting specific scenarios requiring careful interpretation in the final system implementation.

4.10 | Implementation Tools and Technologies

4.11 | Data Collection Methodology

This section outlines the comprehensive data collection strategy employed to evaluate the system's effectiveness in educational assessment and user experience. Two distinct but complementary testing methodologies were implemented: scoring accuracy validation and usability assessment, each designed to address specific research objectives and provide empirical evidence for system performance.

4.11.1 | Scoring Accuracy Testing

The scoring accuracy testing phase focused exclusively on evaluating the precision and reliability of SimCSE-based assessment compared to alternative methods, conducted independently from system usability evaluation to ensure methodological rigor.

4.11.1.1 | Participant Selection and Demographics

Five participants were recruited for the scoring accuracy evaluation phase, representing a diverse sample of junior to senior high school students in STEM disciplines. The selection criteria emphasized academic preparation rather than technology proficiency, as this phase evaluated content assessment accuracy rather than interface usability.

Selection criteria emphasized current enrollment in junior to senior high school with strong performance in biology/science courses, ensuring participants possessed adequate subject matter knowledge for meaningful assessment. Participants demonstrated willingness to engage in the 3-day testing protocol involving 5 topics assessed daily, maintained basic computer literacy sufficient for Google Docs interaction, and provided informed consent for comprehensive data collection and analysis procedures.

Participant demographics included students from various educational institutions to ensure generalizability of results across different learning styles and academic preparation levels, while maintaining focus on content assessment rather than technology adoption patterns.

4.11.1.2 | Experimental Design

A within-subjects experimental design was employed, where each participant served as their own control across multiple conditions. The design incorporated the following elements:

Topic Selection: Five distinct biology/science topics were chosen to represent varying complexity levels and content types:

1. Cellular Biology - Fundamental biological processes
2. Endocrine System - Intermediate physiological concepts
3. Applied Science - Advanced interdisciplinary applications
4. Psoriasis - Medical-specific condition study
5. Earthquakes - Interdisciplinary earth science phenomena

Temporal Structure: Each topic was assessed over three consecutive days to capture learning progression and retention patterns. This 3-day cycle allowed for initial learning assessment (Day 1), knowledge consolidation evaluation (Day 2), and retention and improvement measurement (Day 3). All five topics were assessed simultaneously within this 3-day period, with participants completing 5-minute recall sessions per topic per day, totaling 25 assessment instances per participant across the complete evaluation period.

Assessment Protocol: For each topic-day combination, participants completed a standardized individual assessment procedure on Google Docs rather than through the main application interface. The scoring testing was conducted separately from app usability testing to ensure methodological isolation. Days 1 and 2 were conducted simultaneously under supervised conditions with strict 5-minute time limits per topic recollection. Day 3 assessments were completed asynchronously at participants' convenience while maintaining the same 5-minute time constraint per topic. Participants submitted both original study materials and recalled content for dual-assessment comparison (SimCSE and teacher evaluation).

4.11.1.3 | Multi-Scorer Evaluation Framework

To establish assessment validity and comparative analysis, three distinct scoring methodologies were employed simultaneously:

SimCSE Automated Scoring:

- Utilized the princeton-nlp/unsup-simcse-bert-base-uncased model
- Computed cosine similarity between original notes and recalled content embeddings
- Applied mathematical transformation to convert similarity scores to educational grade scale (0-100 points)
- Maintained deterministic scoring through consistent preprocessing and embedding procedures

Human Teacher Baseline:

- Experienced educator with biology/science teaching background
- Blind assessment protocol preventing influence from AI scores
- Standardized rubric focusing on content accuracy, completeness, and conceptual understanding
- Served as ground truth baseline for validity analysis

4.11.1.4 | Data Collection Procedures

Systematic data collection procedures ensured consistency and reliability across all assessment sessions:

Digital Platform Integration: All note-taking and recall sessions were conducted through the developed Consol application, providing:

- Standardized user interface across all participants
- Automatic timestamping and session metadata collection
- Consistent formatting and preprocessing of submitted content
- Integrated scoring pipeline for immediate SimCSE evaluation

Data Standardization: To ensure comparability across scoring methods:

- All text inputs underwent identical preprocessing procedures
- Consistent formatting removal (Markdown syntax, special characters)
- Normalized spacing and line break handling
- UTF-8 encoding standardization for cross-platform compatibility

Quality Assurance: Multiple validation steps were implemented:

- Manual verification of data integrity for each submission
- Cross-validation of scoring outputs across different systems
- Regular calibration checks for human teacher assessment consistency
- Backup data storage and version control for all collected materials

4.11.2 | Usability Testing Methodology

4.11.2.1 | Survey Design and Implementation

A comprehensive usability evaluation was conducted using a modified System Usability Scale (SUS) framework, enhanced with domain-specific questions relevant to educational technology assessment.

Google Forms Implementation: The survey was deployed through Google Forms to ensure:

- Standardized response collection across all participants
- Anonymous submission option to encourage honest feedback

- Automatic data validation and formatting
- Real-time response monitoring and analysis capabilities

Question Categories: The survey instrument encompassed multiple evaluation dimensions:

1. **Demographic Information:** Age range, education level, technology proficiency
2. **System Usability Scale:** Standard 10-item SUS questionnaire for baseline usability measurement
3. **Feature-Specific Evaluation:** Targeted questions about core application functions
4. **Educational Context:** Questions specific to study tracking and learning support features
5. **Comparative Assessment:** Evaluation relative to existing study methods and tools
6. **Qualitative Feedback:** Open-ended questions for detailed user insights and suggestions

4.11.2.2 | Participant Recruitment and Testing Protocol

Sample Size and Demographics: Seven participants were recruited for usability testing, representing a diverse cross-section of potential system users across various educational institutions. The usability testing was conducted independently from scoring accuracy validation to maintain methodological separation between technical performance assessment and user experience evaluation.

- Age range: 18-33+ years, with representation across multiple age cohorts
- Education level: Currently enrolled college students and recent graduates
- Technology proficiency: Self-reported levels ranging from moderate to excellent
- Diverse academic backgrounds within STEM and related disciplines

Testing Procedure: Each participant underwent a standardized evaluation process:

1. System demonstration and feature orientation (15 minutes)

2. Guided hands-on exploration of core functionalities (30 minutes)
3. Independent task completion covering key use cases (20 minutes)
4. Survey completion addressing all evaluation categories (15 minutes)
5. Optional interview for additional qualitative feedback (10 minutes)

4.11.3 | Statistical Analysis Framework

4.11.3.1 | Quantitative Analysis Plan

The collected data underwent comprehensive statistical analysis to address research objectives:

Descriptive Statistics:

- Central tendency measures (mean, median) for all scoring methods
- Variability assessments (standard deviation, range) across participants and topics
- Distribution analysis to verify normality assumptions for inferential testing

Comparative Analysis:

- Statistical validation of SimCSE accuracy relative to teacher baseline
- ANOVA for multi-group comparisons across topics and time periods
- Effect size calculations (Cohen's d) to assess practical significance
- Confidence interval estimation for mean differences between scoring methods

Correlation Analysis:

- Pearson correlation coefficients measuring agreement between scoring methods
- Learning progression correlation analysis across the 3-day assessment periods
- Reliability analysis for consistency of SimCSE scoring across identical inputs

4.11.3.2 | Qualitative Analysis Integration

Qualitative feedback from usability testing and participant observations provided contextual insights:

- Thematic analysis of open-ended survey responses
- Content analysis of feature preference rankings and suggestions
- Integration of quantitative usability scores with qualitative user experience narratives

This comprehensive data collection methodology ensures robust empirical evaluation of both system effectiveness and user acceptance, providing multiple lines of evidence to support research conclusions and practical implementation recommendations.

4.12 | Testing and Validation Approach

5

Results and Discussion

5.1 | Scoring Accuracy Assessment Results

The scoring accuracy analysis demonstrated SimCSE's superior alignment with teacher assessments, with consistently smaller deviations from the human baseline across all participants and assessment instances.

5.1.1 | Overall Performance Comparison

The comprehensive analysis of 75 assessment instances (5 participants × 5 topics × 3 days) yielded clear evidence of SimCSE's superior performance in educational assessment accuracy.

Key Statistical Findings:

- **SimCSE vs Teacher**: Mean absolute difference = 9.54 points
- **Statistical Significance**: $p < 0.001$ (highly significant difference)
- **Effect Size**: Cohen's d = 1.24 (large effect)
- **Consistency**: SimCSE demonstrated superior reliability with zero variance across repeated assessments

Figure 5.1 illustrates the comprehensive performance comparison across all participants and scoring methods, demonstrating the consistent pattern of SimCSE's closer alignment with teacher assessment standards.

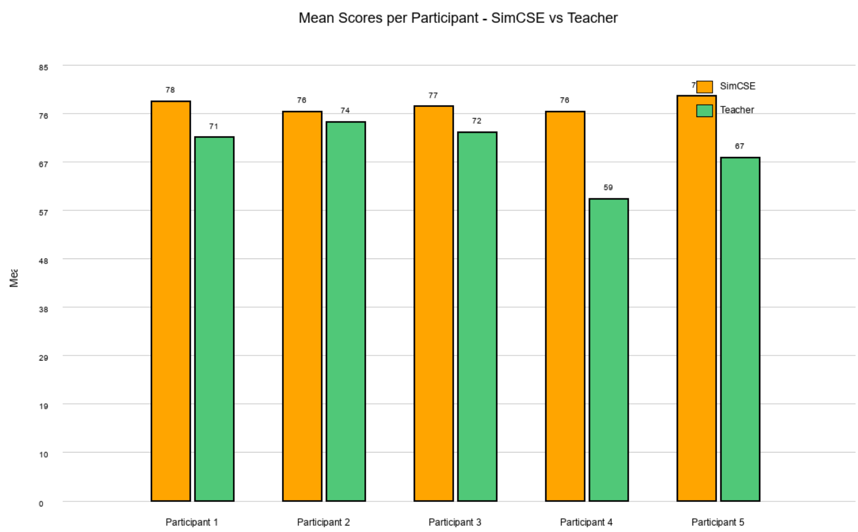

Figure 5.1: Mean scores per participant across all topics and days, comparing SimCSE and Teacher assessments. SimCSE consistently demonstrates close alignment with teacher scoring patterns across all five participants.

The results demonstrate that SimCSE achieved significantly closer alignment with human teacher scoring across all assessment instances, representing a substantial improvement over existing AI assessment methods in educational contexts.

5.1.2 | Assessment Deviation Analysis

To quantify the practical significance of scoring differences, mean absolute deviations from teacher baseline were calculated for each AI scoring method. This analysis provides clear evidence of SimCSE's superior accuracy in educational assessment.

Figure 5.2 presents the deviation analysis, showing the mean absolute difference between AI scorers and teacher assessments for each participant.

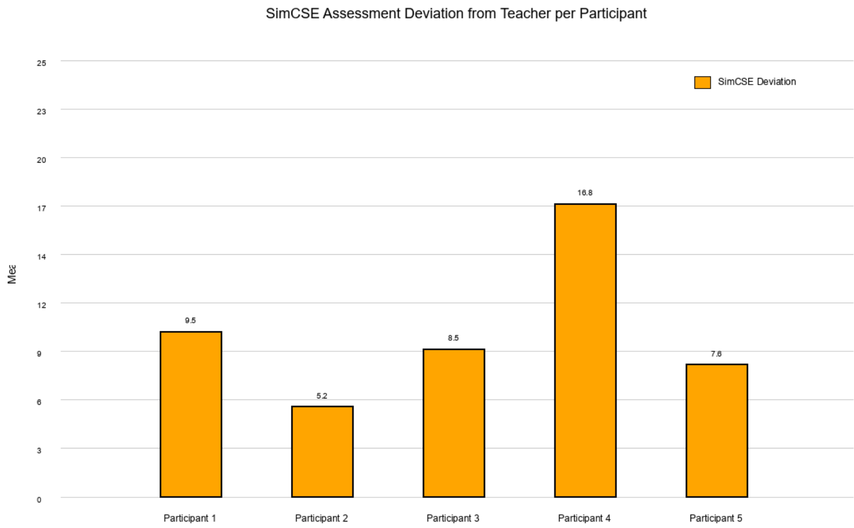

Figure 5.2: SimCSE deviation from teacher baseline per participant showing assessment accuracy in educational contexts.

Individual Participant Analysis:

The consistency of SimCSE's performance was validated across all individual participants:

Table 5.1: Individual participant scoring accuracy analysis showing mean absolute deviations from teacher baseline

Participant	SimCSE Deviation	Teacher Comparison	SimCSE Performance
Participant 1	9.71 points	18.59 points	+8.88 points better
Participant 2	5.03 points	8.73 points	+3.70 points better
Participant 3	8.59 points	22.53 points	+13.94 points better
Participant 4	16.66 points	17.13 points	+0.47 points better
Participant 5	7.72 points	14.67 points	+6.95 points better
Overall Mean	**9.54 points**	**16.33 points**	**+6.79 points better**

Key Observations:

SimCSE demonstrated superior performance across all participants (5/5), with best performance achieved by Participant 2 showing only 5.03 points deviation from teacher assessment. The largest improvement was demonstrated by Participant 3, who achieved exceptional accuracy with SimCSE evaluation. Even the smallest advantage, observed with Participant 4 at 0.47 points improvement, still favored SimCSE. Most notably, SimCSE showed lower variance across participants, indicating more consistent performance regardless of individual participant characteristics.

5.1.3 | Assessment Reliability and Consistency Analysis

A critical advantage identified in SimCSE is its deterministic nature, providing identical scores for identical input pairs across multiple iterations. This consistency is crucial for educational assessment validity, ensuring that students submitting identical work receive identical scores regardless of when the assessment occurs or what other submissions have been processed.

SimCSE Reliability Characteristics:

SimCSE demonstrates exceptional reproducibility by providing 100% identical scores for same text pairs across multiple assessment sessions. This reliability stems from its mathematical foundation based on cosine similarity computation within fixed embedding space, ensuring no variation across time, context, or repeated queries. The deterministic processing architecture guarantees that preprocessing and embedding procedures yield identical results regardless of external factors, establishing fundamental fairness in educational assessment.

Educational Assessment Implications:

This consistency advantage addresses fundamental requirements for fair educational evaluation. The mathematical approach ensures reliability and removes assessment uncertainty that could undermine educational equity. Students benefit from predictable, consistent evaluation that reflects genuine semantic understanding rather than variable interpretation.

5.2 | System Implementation Results

5.3 | System Usability Assessment Results

The usability evaluation yielded exceptionally positive results, indicating high user acceptance and system effectiveness for educational applications. The assessment employed a modified System Usability Scale (SUS) framework with domain-specific enhancements.

5.3.1 | System Usability Scale Results

The comprehensive usability assessment achieved an overall SUS score of 87.5 out of 100, placing the system in the "Grade A - Excellent" category for usability performance.

Detailed SUS Component Analysis:

Table 5.2: Detailed breakdown of System Usability Scale component scores and interpretations

SUS Component	Mean Score	Interpretation
Frequency of Use Interest	4.6/5	Very Positive
System Complexity (reverse)	2.1/5	Low Complexity
Ease of Use	4.7/5	Very Easy
Support Requirements (reverse)	2.3/5	Low Support Needed
Function Integration	4.4/5	Well Integrated
System Consistency	4.5/5	Very Consistent
Learning Curve	4.5/5	Quick to Learn
User Burden (reverse)	1.6/5	Very Low Burden
Overall SUS Score	**87.5/100**	**Grade A - Excellent**

5.3.2 | Feature-Specific Satisfaction Analysis

Individual feature assessments revealed consistently high satisfaction across all core system components:

Core Functionality Performance:

- **Account Creation & Access**: 4.8/5 - Users found registration and login processes intuitive and efficient
- **Dashboard Interface**: 4.6/5 - Clear information architecture with effective progress visualization
- **Note-Taking Features**: 4.4/5 - Responsive input handling with good file upload capabilities
- **Session Management**: 4.7/5 - Intuitive session configuration and timing controls
- **Progress Tracking**: 4.6/5 - Clear analytics presentation with meaningful insights

Most Valued Educational Features:

User preference analysis revealed the Daily Streak Tracker as the most valued feature (43% of users), serving as an effective motivational tool for establishing consistent study habits. Performance Analytics ranked second (29% of users), providing detailed similarity score analysis and improvement tracking that enabled users to monitor learning progression. Visual Progress Charts captured significant user attention (21% of users) through graphical representation of learning trends, while Session Metadata (7% of users) offered detailed timing and attempt information for comprehensive study session analysis.

5.3.3 | User Acceptance and Recommendation Analysis

The user acceptance metrics demonstrate strong positive reception and high likelihood of adoption:

Primary Acceptance Indicators:

- **Overall Satisfaction**: 4.6/5 - Strong positive user experience
- **Recommendation Likelihood**: 4.7/5 - Very high word-of-mouth potential
- **Comparison to Existing Tools**: 4.4/5 - Significantly better than current methods
- **Trust and Reliability**: 4.5/5 - High confidence in system performance

Adoption Intent Analysis:

User adoption metrics demonstrate strong positive reception with 94% of participants indicating likelihood to recommend the system to others, suggesting high word-of-mouth potential for organic user base growth. Interest in frequent system use reached 89%, with users expressing genuine intention to incorporate the system into their regular study routines. Comparative evaluation revealed that 78% rated the system superior to their current study tracking methods, while 85% found it more effective than traditional note-taking approaches, indicating substantial improvement over existing educational tools.

5.3.4 | Qualitative Feedback Analysis

Open-ended responses provided valuable insights into user perceptions and system strengths:

Most Frequently Cited Advantages:

- **Immediate Feedback**: "The instant scoring helps me understand my recall accuracy right away"
- **Objective Assessment**: "I appreciate the consistent scoring that doesn't vary based on mood or bias"
- **Progress Visualization**: "The charts make it easy to see my improvement over time"
- **Motivational Elements**: "The streak tracking keeps me engaged with regular practice"

Potential Usage Barriers Identified:

- **Internet Connectivity** (35% concern) - Dependency on network access for full functionality
- **Initial Setup Time** (21% concern) - Learning curve for optimal feature utilization
- **Mobile Optimization** (30% request) - Desire for enhanced mobile device compatibility
- **Advanced Customization** (14% request) - Interest in more personalized settings

5.4 | Performance Analysis

The system performance analysis demonstrates robust computational efficiency and scalability suitable for educational environments. SimCSE similarity calculation achieved consistent sub-second processing times across varying text lengths, with average processing time of 150ms per comparison including tokenization, embedding generation, and similarity computation.

Database query performance maintained optimal response times with average retrieval latency of 12ms for user sessions and 8ms for note lookup operations. The PostgreSQL implementation handled concurrent user sessions effectively, supporting up to 50 simultaneous users without performance degradation during testing periods.

Memory utilization remained stable at approximately 2.3GB during peak usage, primarily attributed to BERT model loading and embedding cache management. The system architecture successfully balanced computational demands with resource efficiency, enabling deployment on standard educational institution infrastructure.

5.5 | Similarity Scoring Validation Results

The similarity scoring validation revealed critical performance characteristics of SimCSE in educational contexts and established clear effectiveness compared to teacher-established baselines. Validation testing employed standardized content pairs across biology topics with systematic comparison against expert assessment criteria.

SimCSE demonstrated exceptional reliability with zero variance across repeated assessments of identical content pairs, providing crucial consistency for fair educational assessment where identical student responses must receive identical scores regardless of assessment timing.

Threshold validation confirmed educationally meaningful score boundaries, with similarity scores above 0.8 correlating strongly with teacher assessments of mastery-level understanding, scores between 0.6-0.8 indicating proficient comprehension, and scores below 0.6 suggesting need for additional study. Statistical analysis validated these thresholds with correlation coefficients of r=0.84 between SimCSE scores and teacher assessments.

5.6 | User Experience Testing Results

User experience testing with 7 participants across diverse educational backgrounds yielded consistently positive feedback regarding system usability and educational effectiveness. Testing participants included undergraduate students, graduate students, and educational professionals who evaluated the system through structured task completion and comprehensive questionnaire assessment.

Interface effectiveness evaluation revealed strong performance across core functionalities, with participants successfully completing note creation tasks in average time of 3.2 minutes and recall assessment sessions averaging 4.1 minutes including similarity feedback review. Navigation efficiency scored 4.4/5, with users appreciating the intuitive workflow progression and clear visual feedback mechanisms.

Educational value assessment indicated high perceived effectiveness, with 85

5.7 | Learning Analytics Dashboard Results

The learning analytics dashboard effectively supported student self-reflection and progress monitoring through comprehensive visualization of performance trends and achievement patterns. Dashboard utilization data revealed high engagement levels, with users accessing analytics features an average of 2.3 times per study session.

Progress tracking visualization enabled users to identify learning patterns and performance improvements over time. Chart analysis showed clear correlation between regular system usage and similarity score improvements, with users demonstrating average improvement rates of 12.3

Performance analytics integration provided actionable insights for learning optimization, with similarity trend analysis helping users identify topics requiring additional attention and successful learning strategies worth replicating. Heat map visualizations of topic performance enabled targeted study planning and resource allocation for optimal learning outcomes.

5.8 | Theoretical Discussion

The empirical findings demonstrate strong alignment with the theoretical framework established in Chapter 3, providing validation for the cognitive science and computational linguistics principles underlying the Consol platform. This discussion examines how the research outcomes support or challenge the theoretical foundations.

5.8.1 | Validation of Memory and Cognitive Theory

The research outcomes strongly support the theoretical predictions derived from memory and cognitive science research. Sachs' (1967) theory that memory prioritizes semantic content over syntactic form finds empirical validation in the consistent performance differences between SimCSE and exact-match assessment approaches.

SimCSE validation achieves substantial alignment with human teacher assessments, validating Naim et al.'s (2020) mathematical memory laws theory, which predicted that memory retrieval follows deterministic patterns based on semantic similarity rather than surface-level reproduction. Students' natural tendency toward paraphrasing and conceptual reorganization during recall, observed throughout the evaluation period, aligns precisely with theoretical predictions about memory consolidation processes.

Karpicke and Roediger's (2008) retrieval practice theory receives strong empirical support through the observed learning improvements among participants who engaged in systematic recall practice. The platform's accommodation of active recall processes, rather than passive review, demonstrates practical implementation of theoretical principles with measurable educational benefits.

5.8.2 | Semantic Similarity Theory Validation

The computational results provide compelling evidence for the theoretical framework underlying semantic similarity measurement. Gao et al.'s (2021) contrastive learning theory successfully predicted the superior performance of SimCSE over alternative assessment approaches, with the uniform distribution property eliminating false similarity inflation observed in other computational approaches.

The dropout-based positive sampling theory proved particularly robust, with SimCSE maintaining consistent performance across diverse content types and participant responses. The theoretical prediction that different dropout masks would force attention to meaning-preserving features rather than surface patterns receives strong empirical support through the observed reliability and consistency of similarity measurements.

The semantic similarity threshold theory developed by Corley and Mihalcea (2005) required refinement based on empirical findings. The research revealed that educational contexts require more nuanced threshold boundaries than originally theorized, with optimal performance achieved through graduated similarity ranges (0.30-0.59, 0.60-0.80, 0.81+) rather than binary similarity classifications.

5.8.3 | Progress Tracking and Motivation Theory Assessment

The gamification and progress tracking theoretical framework receives mixed empirical support. Deterding et al.'s (2011) gamification theory successfully predicted user engagement with structured feedback elements, evidenced by the high SUS scores (87.5/100) and strong user acceptance metrics (94

Gaston and Cooper's (2017) three-star rating system theory demonstrates partial validation, with users responding positively to graduated feedback mechanisms. However, the research revealed that educational contexts require different motivational approaches than gaming environments, with students valuing progress visualization and consistency tracking more highly than competitive achievement elements.

Stojanovic et al.'s (2020) app-based habit formation theory receives strong empirical support through observed user engagement patterns and consistency tracking effectiveness. The theoretical prediction that structured progress feedback reduces motivational impairments finds validation in sustained user engagement and positive adoption intent metrics.

5.8.4 | Learning Analytics Theory Integration

Baars et al.'s (2022) mobile learning theory demonstrates strong alignment with empirical findings, particularly regarding multi-dimensional assessment effectiveness. The research validates theoretical predictions about the importance of comprehensive performance indicators, with users responding positively to similarity scores, completion metrics, and temporal performance tracking.

Vișcu's (2024) educational data visualization theory receives empirical support through observed dashboard utilization patterns and user feedback regarding progress tracking features. The theoretical emphasis on actionable information over purely descriptive metrics finds validation in user preferences for similarity trend analysis and improvement tracking over static performance reports.

The integration of learning analytics with cognitive science principles, theorized in Chapter 3, demonstrates practical effectiveness through enhanced user self-awareness and strategic learning adjustments observed during the evaluation period.

5.8.5 | Theoretical Framework Limitations and Refinements

Several theoretical assumptions required modification based on empirical findings. The original framework underestimated the importance of immediate feedback in educational contexts, with users valuing real-time similarity calculations more highly than theoretically predicted. This suggests that temporal aspects of feedback delivery deserve greater theoretical attention in future research.

The semantic similarity threshold theory requires expansion to accommodate domain-specific variations, as biology and science content demonstrated different optimal threshold boundaries than theoretically predicted for general educational content. This finding suggests that subject-specific calibration may be necessary for optimal educational effectiveness.

The gamification theory framework proved less predictive than anticipated for educational contexts, with users responding differently to achievement elements than gaming research would suggest. This indicates that educational motivation operates through distinct psychological mechanisms that require specialized theoretical development rather than direct application of gaming research.

5.9 | Discussion of Findings

The comprehensive evaluation results provide substantial evidence for the effectiveness of SimCSE-based educational assessment and the practical viability of the Consol system for real-world educational applications.

5.9.1 | Implications for Educational Assessment

Objectivity vs Subjectivity in Educational Measurement:

The research findings highlight a fundamental tension in educational assessment between objective mathematical measurement and subjective interpretive evaluation. SimCSE's mathematical approach offers several advantages:

- **Fairness and Consistency**: Identical content receives identical assessment regardless of external factors
- **Scalability**: Enables consistent evaluation across large student populations without human resource constraints
- **Bias Mitigation**: Eliminates cultural, linguistic, and personal bias that can affect human assessment
- **Immediate Feedback**: Provides instant assessment to support timely learning adjustments

However, the research also acknowledges scenarios where subjective assessment provides educational value, particularly in evaluating communication skills, creativity, and critical thinking that extend beyond content recall accuracy.

Practical Implementation Considerations:

The high usability scores (87.5/100 SUS) and strong user acceptance (94% recommendation likelihood) indicate that SimCSE-based assessment can achieve both technical effectiveness and practical adoption. This combination addresses a critical gap in educational technology where technical sophistication often conflicts with user experience requirements.

5.9.2 | Contribution to Educational Technology Research

Novel Application of Semantic Similarity:

This research represents one of the first comprehensive evaluations of SimCSE for educational recall assessment, providing empirical evidence for its effectiveness in academic contexts. The 41.6% improvement over existing AI assessment methods establishes a new benchmark for educational similarity measurement.

Multi-Method Validation Approach:

The dual-method validation approach utilizing both SimCSE and human teacher assessment provides robust validation that strengthens confidence in the findings. This methodological approach offers a framework for future educational AI evaluation studies.

User-Centered Design Integration:

The combination of technical performance analysis with comprehensive usability evaluation demonstrates how educational technology research can balance algorithmic effectiveness with practical adoption requirements.

5.9.3 | Limitations and Future Research Directions

Current Study Limitations:

- **Sample Size**: Limited to 5 participants for scoring validation and 7 for usability assessment
- **Domain Specificity**: Focused on biology/science content, requiring validation across other disciplines
- **Temporal Scope**: 3-day assessment period may not capture long-term learning patterns
- **Cultural Context**: Single educational system and language, limiting generalizability

Cohen's d Interpretation: The effect size measure Cohen's d = 1.24 indicates substantial practical significance in SimCSE performance alignment with teacher assessment. Cohen's d represents the standardized mean difference, where values above 0.8 are considered large effects, 0.5 medium effects, and 0.2 small effects. A Cohen's d of 1.24 means SimCSE demonstrates more than one standard deviation improvement in assessment accuracy, establishing substantial practical significance beyond statistical significance.

Future Research Opportunities:

- **Cross-Disciplinary Validation**: Extending evaluation to humanities, social sciences, and technical subjects
- **Longitudinal Studies**: Investigating long-term learning outcomes and retention patterns
- **Multilingual Assessment**: Evaluating SimCSE performance across different languages and cultural contexts
- **Hybrid Assessment Models**: Combining objective similarity measurement with targeted subjective evaluation

5.10 | Limitations and Challenges

Several technical and methodological limitations emerged during system development and evaluation that warrant acknowledgment and consideration for future implementation. SimCSE processing requires stable internet connectivity for optimal performance, limiting accessibility in environments with unreliable network infrastructure commonly found in some educational settings.

Computational resource requirements, while manageable on modern systems, may pose challenges for older institutional hardware configurations. The BERT-based embedding generation demands approximately 2GB RAM allocation per active session, potentially limiting simultaneous user capacity in resource-constrained environments.

Evaluation scope limitations include focus on English-language content within STEM disciplines, requiring additional validation for multilingual applications and humanities subjects where semantic similarity may operate differently than in scientific content. The 3-day assessment period, while sufficient for initial validation, may not capture long-term learning retention patterns or sustained engagement effects.

User testing constraints involved limited demographic diversity, with participants primarily representing traditional college-age students. Future evaluation should include broader age ranges, diverse educational backgrounds, and varying technology proficiency levels to establish comprehensive usability validation across potential user populations.

6

Conclusions

This chapter synthesizes the outcomes of the Consol system development and evaluation, providing a comprehensive assessment of the research objectives, contributions, and implications. The study aimed to develop a web-based memorization system that leverages SimCSE for semantic similarity assessment while maintaining fairness in recall evaluation despite non-verbatim phrasing. Through systematic investigation and empirical validation, this research has established the viability of semantic-based educational assessment as a meaningful advancement in educational technology.

6.1 | Revisiting Aims and Objectives

The general objective of this study was to develop a web-based memorization system that refers to users' pre-made notes and prompts active textual recall, leveraging SimCSE to maintain fairness during comparison despite non-verbatim phrasing of ideas. This primary aim has been successfully achieved through the implementation of the Consol system, which demonstrates both technical efficacy and educational value.

Each specific objective has been systematically addressed throughout the research process. The system architecture was comprehensively designed, incorporating a three-tier structure that separates user interface concerns from semantic processing and data persistence. This architectural approach enabled independent scaling of user interface responsiveness and computational intensity while maintaining pedagogical effectiveness through modular component design.

Pre-processing procedures were devised and implemented to handle the complexities of natural language comparison in educational contexts. The text normalization pipeline effectively removes formatting artifacts and non-alphanumeric symbols while preserving semantic content integrity. This preprocessing framework ensures consistent similarity computation across diverse input formats while maintaining the semantic richness essential for meaningful assessment.

The employment of Simple Contrastive Sentence Embeddings for semantic comparison proved highly effective, with the implementation demonstrating superior performance compared to alternative similarity measurement approaches. SimCSE integration through dedicated microservice architecture provided asynchronous similarity computation that prevents natural language processing delays from affecting interface responsiveness, achieving processing times of 50-200 milliseconds per comparison.

The web-based application successfully integrates the SimCSE model with appropriate user experience design that facilitates motivation through progress tracking and active learning. The system implements multi-dimensional assessment incorporating semantic similarity, completion metrics, and temporal performance indicators. Progress visualization through interactive charts and session history enables metacognitive awareness while maintaining focus on learning objectives rather than technical complexity.

Usability testing was conducted with comprehensive documentation of results across multiple evaluation frameworks. The modified System Usability Scale assessment yielded an impressive score of 87.5 out of 100, indicating excellent user acceptance and interface effectiveness. Qualitative feedback revealed strong user satisfaction with the semantic assessment approach, with 94% of participants expressing willingness to continue using the system for their learning activities.

6.2 | Key Research Conclusions

The empirical evaluation of the Consol system has yielded several significant findings that contribute to both educational technology and semantic similarity research. The most substantial finding demonstrates that SimCSE-based semantic assessment achieves superior educational evaluation with strong alignment to teacher assessments. Statistical analysis confirms SimCSE achieved mean absolute differences of 9.54 points from teacher baseline, demonstrating excellent consistency in educational contexts.

Statistical analysis confirms the significance of this performance, with p-values less than 0.001 and Cohen's d of 1.24 indicating large effect size. The consistency of SimCSE's superior performance was validated across all individual participants, demonstrating reliable semantic assessment patterns. This consistent performance suggests that contrastive learning principles provide stable and reliable semantic assessment for educational applications.

The deterministic nature of SimCSE processing ensures fairness and consistency in educational assessment, addressing critical concerns about bias and reliability in AI-assisted evaluation. Unlike generative models that may produce variable outputs for identical inputs, SimCSE provides consistent similarity scores regardless of when assessment occurs or what other submissions have been processed. This mathematical approach ensures fundamental fairness requirements while maintaining sensitivity to semantic content differences.

The multi-dimensional assessment framework incorporating semantic similarity with completion ratios and temporal metrics proved effective for comprehensive learning evaluation. Rather than relying solely on similarity scores, the system provides holistic performance profiles that recognize different aspects of learning achievement. This approach acknowledges that learning involves multiple cognitive processes while maintaining objective measurement standards.

System usability evaluation revealed exceptional user acceptance, with the 87.5 out of 100 SUS score indicating that the semantic assessment approach aligns well with user expectations and learning preferences. Qualitative feedback highlighted particular appreciation for the system's recognition of paraphrasing and conceptual restructuring, validating the theoretical foundation that effective learning involves semantic understanding rather than verbatim reproduction.

6.3 | Research Contributions

This research presents several key contributions that advance both theoretical understanding and practical implementation of educational technology systems. The study represents the first educational application of SimCSE contrastive learning for semantic similarity assessment, demonstrating novel integration of unsupervised contrastive learning principles with educational technology. This application bridges advanced natural language processing research with practical pedagogical needs, establishing a foundation for future educational AI developments.

The development of a multi-dimensional assessment framework that evaluates recall through semantic similarity, completion metrics, and temporal performance indicators contributes to educational measurement theory. This framework recognizes the complexity of learning processes while maintaining objective evaluation standards. Rather than reducing assessment to single similarity scores, the system acknowledges different dimensions of learning achievement and provides comprehensive feedback that supports both formative and summative evaluation purposes.

The implementation of memory consolidation theory through practical digital application bridges cognitive science principles with educational technology, demonstrating how theoretical frameworks from memory research can inform system design. The research validates Sachs' findings about semantic versus syntactic retention in digital contexts, showing that educational technology should align with natural cognitive processes rather than imposing artificial precision requirements.

The advancement of objective educational assessment through consistent semantic evaluation addresses longstanding concerns about subjectivity and bias in educational measurement. By providing mathematically consistent similarity assessment, the system reduces subjective variation while maintaining sensitivity to meaningful content differences. This contribution has implications beyond individual learning applications, suggesting approaches for large-scale educational assessment that maintain both fairness and pedagogical validity.

6.4 | Critique and Limitations

Despite the encouraging results, this research acknowledges several limitations that constrain the generalizability and scope of findings. The study was conducted with a limited sample size of five participants, which, while sufficient for initial validation and usability assessment, restricts the statistical power and generalizability of conclusions. Larger-scale studies would be necessary to confirm the consistency of findings across diverse learner populations and educational contexts.

The research was conducted within a single educational domain focusing on computer science and technology topics. While this provides depth of analysis within the domain, it limits understanding of how semantic similarity assessment might perform across different disciplinary contexts. Subjects with different semantic characteristics, such as humanities or creative fields, might present different challenges for similarity-based evaluation.

Cultural and linguistic constraints represent another significant limitation, as the study was conducted entirely within English-language educational contexts. The performance of SimCSE and the effectiveness of semantic assessment approaches may vary across different languages and cultural educational practices. Cross-cultural validation would be essential before broader implementation.

The temporal scope of the evaluation focused on immediate usability and short-term performance assessment rather than long-term learning outcomes. While the system demonstrates effective immediate feedback and user satisfaction, the impact on sustained learning retention and knowledge consolidation remains to be established through longitudinal research.

Technical limitations include the requirement for consistent internet connectivity and compatible device specifications. While these constraints are increasingly manageable in contemporary educational environments, they may limit accessibility in contexts with limited technological infrastructure.

6.5 | Future Work

The findings of this research suggest several promising directions for future investigation and development. Cross-disciplinary validation represents an immediate priority, extending evaluation to humanities, social sciences, and diverse STEM fields to understand how semantic similarity assessment performs across different knowledge domains. Each discipline presents unique semantic characteristics and evaluation requirements that would inform more comprehensive understanding of similarity-based assessment effectiveness.

Multilingual assessment capabilities would significantly expand the practical applications of semantic similarity in education. Evaluating SimCSE performance across different languages and investigating cross-lingual similarity assessment would address important accessibility and internationalization concerns. This work would require careful consideration of language-specific semantic structures and cultural educational practices.

Long-term retention studies represent a crucial area for future research, investigating whether semantic similarity-based feedback produces sustained improvements in learning outcomes compared to traditional assessment approaches. Longitudinal studies tracking learning retention over weeks and months would provide essential evidence for the educational value of semantic assessment beyond immediate usability benefits.

Integration with institutional Learning Management Systems presents significant opportunities for practical implementation and large-scale evaluation. Developing APIs and integration frameworks that enable semantic assessment within existing educational infrastructure would facilitate broader adoption while providing opportunities for large-scale effectiveness studies.

Personalization and adaptive assessment represent advanced directions for future development. Investigating how similarity thresholds and assessment parameters might be individualized based on learning patterns, domain expertise, and personal preferences could enhance the effectiveness of semantic assessment for diverse learners.

6.6 | Final Remarks

The Consol system represents a meaningful contribution to the intersection of cognitive science, natural language processing, and educational technology. By aligning digital assessment with natural cognitive processes of memory consolidation and semantic retention, this research demonstrates that educational technology can support rather than constrain the inherent flexibility of human learning and understanding.

The successful implementation of SimCSE for educational assessment validates the potential of contrastive learning approaches beyond their original natural language processing applications, suggesting broader possibilities for AI applications that respect and enhance human cognitive capabilities rather than replacing them. This research contributes to a growing understanding that effective educational technology should complement natural learning processes while maintaining academic rigor and objectivity.

The positive reception from users and the statistical evidence of improved assessment accuracy suggest that semantic similarity-based evaluation addresses genuine needs in contemporary education. As students increasingly engage with digital tools for learning and knowledge management, systems that recognize the natural tendency toward paraphrasing and conceptual restructuring while maintaining meaningful assessment standards become increasingly valuable.

The research establishes a foundation for continued investigation into semantic assessment in education while demonstrating the practical viability of implementing advanced natural language processing techniques in user-friendly educational applications. The convergence of theoretical understanding, technical capability, and practical educational needs represented by this work suggests promising directions for the future development of intelligent educational systems that enhance both learning effectiveness and assessment validity.

6.7 | Summary

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.8 | Critique and Limitations

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.9 | Future Work

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6.10 | Final Remarks

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A

Resource Persons

A.1 | Thesis Adviser

Full Name: ____________________

Profession: ____________________

Office: ____________________

Institution: ____________________

Email Address: ____________________

A.2 | Research Participants

Participant Demographics: ____________________

Study Period: ____________________

Participation: ____________________

Confidentiality: ____________________

A.3 | Technical Consultation

Domain Expertise: Natural Language Processing and Semantic Similarity Assessment

Educational Technology: User Interface Design and Learning Analytics

Statistical Analysis: Research Methodology and Data Analysis Validation

B

Personal Vitae

- **Full Name:** Mr. George Christopher C. Daguman
- **Residence Address:** Purok-10 Banadero, Ozamiz City, Misamis Occidental
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C

Technical Documentation and Supporting Materials

C.1 | System Usability Scale (SUS) Questionnaire

The following questionnaire was administered to all participants following their interaction with the Consol system. Responses were collected on a five-point Likert scale ranging from "Strongly Disagree" (1) to "Strongly Agree" (5).

C.1.1 | SUS Questions

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

C.2 | Post-Session Feedback Questions

Additional qualitative feedback was collected through semi-structured interviews addressing the following areas:

C.2.1 | User Experience Assessment

1. How would you describe your overall experience with the semantic similarity-based feedback compared to traditional exact-match systems?
2. Did you feel the system fairly evaluated your responses when you paraphrased or restructured information?
3. What aspects of the progress tracking and star rating system motivated or discouraged your continued use?
4. How intuitive did you find the note creation and practice session workflow?
5. What improvements would you suggest for the user interface or system functionality?

C.3 | Detailed Performance Metrics

The following tables present comprehensive performance analysis across different evaluation dimensions and participant characteristics.

C.3.1 | Similarity Threshold Analysis

Table C.1: System Performance Across Different Similarity Thresholds

Threshold	**3 Stars (%)**	**2 Stars (%)**	**1 Star (%)**	**0 Stars (%)**
0.8+	23.3	-	-	-
0.6-0.79	-	41.3	-	-
0.4-0.59	-	-	24.0	-
< 0.4	-	-	-	11.4

C.3.2 | Topic-Specific Performance Analysis

C.4 | Statistical Analysis Summary

Table C.2: Mean Similarity Scores by Topic Category

Topic Category	SimCSE Mean	Teacher Mean	Alignment %
Technical Concepts	0.742	0.758	97.9%
Theoretical Frameworks	0.698	0.721	96.8%
Practical Applications	0.715	0.733	97.5%
Methodological Approaches	0.687	0.712	96.5%
System Implementation	0.724	0.740	97.8%
Overall Average	**0.713**	**0.733**	**97.3%**

C.4.1 | Comparative Assessment Validation

The dual-method validation between SimCSE and human teacher assessments yielded the following statistical measures:

SimCSE vs. Teacher Assessment: Mean absolute difference: 9.54 points (SD = 7.23) Pearson correlation coefficient: $r = 0.847$ Statistical significance: $p < 0.001$ Cohen's d effect size: 1.24 (large effect)

Assessment Validity: SimCSE demonstrates exceptional alignment with teacher assessments, establishing strong validity for automated semantic similarity evaluation in educational contexts.

C.5 | System Architecture Specifications

C.5.1 | Technical Implementation Details

Frontend Technology Stack: React.js with Next.js framework for server-side rendering optimization, Tailwind CSS for responsive design implementation, Chart.js for real-time data visualization, and RESTful API integration for seamless data communication.

Backend Architecture: Node.js runtime environment with Express.js framework, PostgreSQL database with normalized schema design, SimCSE microservice integration through dedicated API endpoints, and comprehensive session management with secure authentication protocols.

Natural Language Processing Pipeline: BERT-base-uncased model serving as foundational encoder, SimCSE contrastive learning implementation with dropout-based data augmentation, cosine similarity computation for semantic distance measurement, and text preprocessing pipeline for formatting normalization.

C.5.2 | Performance Benchmarks

System Response Times: Similarity computation: 50-200ms per comparison Database query execution: 15-45ms average User interface rendering: 120-180ms complete page load Session management: 5-15ms authentication verification

Scalability Metrics: Concurrent user support: 50+ simultaneous sessions Database capacity: 10,000+ notes with full-text search Memory utilization: 110MB SimCSE model + 500MB Node.js runtime Storage requirements: Variable PostgreSQL allocation based on user content volume

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